



FEU Institute of Technology
COLLEGE OF ENGINEERING • COLLEGE OF COMPUTER STUDIES

COMPUTER SCIENCE DEPARTMENT

CS0025 (SOFTWARE ENGINEERING 1)

ACTIVITY 2 [AI – ASSISTED]

RRL AND RRS

Group Name:	Statisfied
Group Member/s:	Dawn Andrei Pamesa (Leader) Rogelio Porras John Ashley Alba Ralph Blanco
Professor:	Doc Beau Gray Habal
Date Performed:	September 21 to 24, 2026
Date Submitted:	September 24, 2026

Assessment Purpose

To revise the feasibility study of each target thesis title from Technical Assessment 1 and support each one with a review of related literature and studies, following the Technical Assessment 2 form: five local literatures, five foreign literatures, five local studies, and five foreign studies per title, each summarised in the group's own words with its relevance, followed by the research and system gap and the refined title. The titles are then compared and ranked by the group, and a primary title and a backup title are identified.

What Changed Since Technical Assessment 1

- The Luzon grid title was dropped, since another group is working on something similar.
- The Barangay Tatalon title was replaced by a child immunization records title at a Mandaluyong City barangay health center.
- Two new partner-based titles were added: SM Foundation, and Quezon City ITDD (carried over and strengthened).
- The retail title was narrowed to milk products with limited shelf life.
- The BAPTC title was kept and its data section updated to the latest reply from BAPTC.

Titles Under Assessment

#	Target Thesis Title	Partner	Primary Technique
1	Highland Vegetable Price and Supply Glut Forecasting at the Benguet Agri-Pinoy Trading Center Using XGBoost	BAPTC and DA-CAR	Predictive Analytics
2	A Data Governance System for Verifiable CSR Impact Reporting Using Fellegi-Sunter Record Linkage	SM Foundation	Data Governance and Ethics
3	A Data Governance System for Privacy Release Readiness of Quezon City Open Data Using Mondrian k-Anonymity	Quezon City ITDD	Data Governance and Ethics
4	Development of a Fuzzy Matching and Data Quality Scoring Model for Detecting Potential Duplicate and Incomplete Child Immunization Records at a Barangay Health Center in Mandaluyong City	Barangay health center, Mandaluyong City	Data Governance and Ethics
5	Demand Forecasting and Dynamic Price Optimization for Milk Retail Products Considering Remaining Shelf Life Using XGBoost	Partner retailer to be confirmed	Predictive Analytics



TARGET TITLE 1 OF 5

Highland Vegetable Price and Supply Glut Forecasting at the Benguet Agri-Pinoy Trading Center Using XGBoost

1. Target Thesis Title

Highland Vegetable Price and Supply Glut Forecasting at the Benguet Agri-Pinoy Trading Center Using XGBoost

Thesis specialization: Data Science

Primary technique: Predictive Analytics, from the approved Data Science technique list

Research domain: Agricultural market forecasting and decision support

Proponents: Dawn Andrei Pamesa (Leader), Rogelio Porras, John Ashley Alba, Ralph Blanco

2. Project Overview

Background of the problem. Benguet's highland farms supply temperate vegetables through the Benguet Agri-Pinoy Trading Center in La Trinidad. BAPTC brings farmers and traders together, but its publicly visible price information is fragmented and mainly backward-looking. Farmers therefore lack a reliable signal for planning harvests before overlapping supply causes sharp price declines.

Specific problem. There is no accessible short-term forecast of wholesale price movement or supply-glut risk by commodity for this market. Planting and harvest decisions are made without any forward indicator, and when many farmers harvest the same commodity in the same window, the resulting oversupply drives prices below recovery cost.

Target users and beneficiaries. The primary users are BAPTC and Department of Agriculture Cordillera Administrative Region staff. Farmers and traders are the intended beneficiaries, reached through existing DA-CAR communication channels rather than through individual accounts in the system.

Proposed solution. A predictive analytics system that forecasts short-term wholesale price movement and supply-glut risk by commodity, delivered as a web dashboard presenting forecasts, risk alerts, model performance, and downloadable summaries.

Main objectives.

- Confirm and obtain structured historical BAPTC price and trading-volume records.
- Use DA Bantay Presyo bulletins as an independent secondary signal for cross-validation.
- Engineer commodity, seasonal, lagged-price, lagged-volume, rolling-trend, and weather features.
- Compare time-series and machine-learning models for price forecasting and glut-risk classification.
- Provide an early-warning dashboard for staff decision support.

Expected output. A cleaned historical price and volume dataset, evaluated forecasting and classification models, a web dashboard with forecasts and risk alerts, and a documented model performance report by commodity.

Intended contribution. An evaluated forecasting method for a Philippine highland-vegetable market that has not yet been modelled in a publicly identified study. The contribution depends on confirming access to suitable BAPTC records.

3. Technical and Technology Feasibility

Area	Proposed Choice	Feasibility and Alternative
------	-----------------	-----------------------------



Programming languages	Python; SQL; JavaScript only if required	Python supports data preparation, forecasting, evaluation, and dashboard services in one stack.
Frameworks and libraries	pandas, NumPy, statsmodels, scikit-learn, XGBoost or LightGBM	All are widely used and open source. Simpler scikit-learn models can replace the boosted libraries if tuning proves difficult.
Database and storage	PostgreSQL or SQLite, with CSV and XLSX import	PostgreSQL fits a shared deployment; SQLite supports an offline prototype with no server administration.
APIs and third-party services	None on the critical path. An API connection is used only if BAPTC exposes one.	This deliberately avoids making an undocumented API a critical dependency.
Platform	Streamlit, or a FastAPI web service with a browser dashboard	A Streamlit prototype reduces development time; FastAPI offers more control if integration is later required.
Hardware and equipment	Development laptops, with optional free-tier cloud hosting	No sensors or specialised equipment are required. Gradient boosting at this data scale runs comfortably on a student laptop.
Architecture and integration	File-based import first, then staging database, then feature engineering, then model training and evaluation, then dashboard	File import keeps the pipeline working regardless of whether BAPTC ever publishes a machine-readable feed.
Security and maintenance	Role-based access, encrypted transport, validated uploads, database backups, and audit logs. Commodity-level public outputs are kept separate from any restricted raw extract.	If the group's web-development capacity is thin, the first version uses Streamlit, SQLite, and scheduled file imports instead of a custom service.
Technical expertise	Python, pandas, and dashboard work are all in hand. Time-series validation discipline is the least familiar area.	Rolling-origin validation is a procedural requirement rather than an advanced technique, and is documented in standard forecasting references.
Limitations and alternatives	The application stack is not the constraint. Historical data availability and consistency are.	DA Bantay Presyo retail bulletins provide an independently available secondary series if BAPTC records are delayed.

Technical finding. The software stack is feasible with open-source tools and the group has the required skills. The main technical constraint is not the application stack but the availability and consistency of the historical data, treated in section 4.

4. Dataset and Data Feasibility

Requirement	Current Assessment
Dataset name and description	BAPTC aggregate daily or weekly wholesale price and trading volume by commodity.
Source, owner, and provider	BAPTC and DA-CAR. On September 10, 2026 the BAPTC price page directed the group to send a formal request to baptc.center@bsu.edu.ph, addressed to Dr. Ruth C. Diego, General Manager, and the request was sent. On September 14, 2026 the Office of the General Manager replied that BAPTC does not release raw or source files, and pointed the group to its published price posts, its 2024 and 2025 Annual Accomplishment Reports, and to DA and PSA. BAPTC has acknowledged the request and the group is waiting for further updates, so whether an aggregate extract can be released is still open.
Public link	No downloadable BAPTC dataset or API is confirmed. BAPTC's daily price posts and its Annual Accomplishment Reports are public reference points, not the planned acquisition method.
Data type and file format	Unknown until the data inventory. Preferred formats are CSV, XLSX, or a database export.
Estimated size	Depends on coverage and reporting frequency. The record count must clear the 10,000-row course floor and is unverified.
Features and attributes	Commodity, date, unit, wholesale price, trading volume, month or season, lagged values, rolling



	statistics, and daily rainfall and temperature from the PAGASA-BSU agrometeorological station in La Trinidad.
Labels and target variables	Short-term price or price change for the regression task, plus a glut-risk label defined by an adviser-approved price-drop threshold for the classification task.
Accessibility and collection	The formal aggregate-data request to the BAPTC General Manager was sent and acknowledged. DA-CAR AMAD (amad@car.da.gov.ph) is a parallel channel and has replied that its concerned section will respond. Weather data is requested from PAGASA through its standard public data request.
Permission and licensing	Document data ownership, allowed uses, retention, publication limits, and whether a data-sharing agreement is required, before any transfer.
Quality, completeness, and bias	Expected issues include missing dates, inconsistent commodity names or units, incomplete volume records, later revisions, and bias toward commodities or periods with better record-keeping.
Preprocessing and storage	Standardise dates, commodities, units, missing-value rules, outliers, and duplicates. Keep raw extracts outside the public repository, record checksums and provenance, restrict access, and publish only approved aggregate outputs.
Training, validation, testing	Chronological rolling-origin validation, never random splitting. A final holdout period stays untouched until model selection is complete.
Alternative data plan	Compile DA Bantay Presyo bulletins as an independently available secondary series, joined with PAGASA-BSU weather records, which are public and do not depend on BAPTC. This supports cross-validation and a narrower price-only study, but cannot replace BAPTC volume data for supply-glut forecasting. Direct Facebook scraping is a last resort requiring adviser approval and a terms-of-service and ethics review.

Dataset finding. Feasibility is conditional. The request reached the right office and was acknowledged, but BAPTC has stated that it does not release raw or source files, so the study depends on BAPTC agreeing to release an aggregate extract instead. No structured record has yet been received. The title should not proceed unchanged unless structured BAPTC records, adequate historical coverage, and permission for research use are all confirmed early in the term; otherwise it narrows to price forecasting on Bantay Presyo and weather data.

5. Algorithm or Model Feasibility

Evaluation Area	Approach 1	Approach 2	Approach 3
Name of algorithm or approach	SARIMA	Prophet	XGBoost gradient-boosted trees
Purpose in the proposed project	Transparent seasonal price baseline	Trend and seasonal forecast with interpretable components	Multivariate price regression and glut-risk classification
Dataset requirements	Regular, sufficiently long univariate series	Regular series; tolerates some missing observations	Lagged price and volume plus calendar and commodity features
Expected performance	Strong when seasonality is stable	Useful baseline; components are easy to explain	Potentially strongest where nonlinear interactions and volume signals matter
Computational requirements	Low	Low to moderate	Moderate; comfortably feasible on student laptops
Advantages	Interpretable with established diagnostics	Fast setup; handles multiple seasonal effects	Flexible; supports mixed features and both regression and classification in one model family
Limitations	Weak against abrupt nonlinear shifts	Can smooth over shocks and is sensitive to tuning	Requires careful leakage control and additional work to explain
Evaluation metrics	MAE, RMSE, MAPE or sMAPE	MAE, RMSE, MAPE or sMAPE	MAE and RMSE for price;



			precision, recall, F1, and PR-AUC for glut risk
Implementation complexity	Moderate	Low to moderate	Moderate

Recommended approach. XGBoost is the primary model, because the study needs one method that handles both the price regression and the glut-risk classification while using lagged volume alongside price, which a univariate seasonal model cannot do. SARIMA is retained as the transparent seasonal baseline and Prophet as a secondary baseline. Model selection uses rolling-origin validation and every candidate is compared against a naive last-value and seasonal-naive forecast, so an apparently good error figure cannot pass unchallenged. If volume data proves unavailable, the study narrows to price forecasting rather than continuing to claim supply-glut prediction.

6. Operational Feasibility

Area	Assessment
Target users and stakeholders	BAPTC and DA-CAR staff are the direct users and the data gatekeepers. Farmers and traders are beneficiaries rather than system users.
Intended operating environment	Office or trading-centre use through a browser on standard computers with a stable internet connection. An offline or local-network deployment supports limited connectivity.
Availability of participants	A BAPTC or DA-CAR contact who can explain the records, authorise access, and review forecast interpretation must be confirmed. Both offices have acknowledged the group's requests, but no named participant has yet agreed.
User needs and benefits	A forward indicator of price movement and oversupply risk by commodity, which does not currently exist for this market in any accessible form.
Devices and facilities	Standard office computers and a browser. No procurement is required.
Usability and accessibility	Dashboard terms, charts, colour contrast, keyboard access, and mobile readability tested with intended staff users.
Training and support	A short user guide and an administrator procedure for importing data, rerunning models, checking failures, and backing up records.
Maintenance	Scheduled model monitoring, because crop patterns, reporting practices, and market behaviour change over time and a model fitted once will drift.
Adoption challenges	The main risk is that forecasts are read as guarantees. Managed by displaying confidence ranges, last-update dates, data gaps, and model limitations rather than presenting single numbers as certainties.

Operational finding. A staff-facing dashboard is practical if BAPTC or DA-CAR participates in data clarification and user testing. Without a partner contact the project can still demonstrate a prototype, but operational validation would be considerably weaker.

7. Economic Feasibility

The amounts below are planning allowances, not supplier quotations. They assume the group already owns suitable laptops and uses open-source software throughout.

Expense	Estimated Cost	Purpose	Possible Alternative
Software licences	PHP 0	Development and analytics tools	Use open-source software
APIs and subscriptions	PHP 0 to 1,200	Optional services or notifications	Use file imports and free tiers
Cloud hosting and storage	PHP 0 to 2,400	Pilot deployment and backups	Local or school-hosted deployment
Dataset acquisition or	PHP 0	Aggregate records and cleaning	Request institutional data; clean



annotation			manually
Hardware or equipment	PHP 0	Development and testing	Use existing laptops
Transportation and data collection	PHP 3,000	BAPTC or DA-CAR visits	Use scheduled calls and email where appropriate, since the request channel is remote
Testing and deployment	PHP 1,000	User testing and incidentals	Use remote testing
Maintenance	PHP 1,200	Backups or post-pilot hosting	Local archive and scheduled exports
Contingency expenses	PHP 1,500	Unexpected project expenses	Reduce optional hosting or travel
Total estimated cost	PHP 6,700 to 10,300	Working budget range	The lower bound uses free tiers throughout

Economic finding. The project can be low-cost because the proposed stack and secondary datasets are free. It carries the highest travel cost among the five titles, driven almost entirely by travel to La Trinidad, which is roughly a five-hour trip from Metro Manila and therefore not a routine visit. The group must confirm its actual budget, travel plan, and any institutional data-access fees before final approval.

8. Schedule Feasibility

Activity	Estimated Duration	Dependency or Control
Requirements gathering	Weeks 1 to 2	Includes the data-access feasibility gate
Literature and related-study review	Weeks 1 to 4	Continues throughout model selection
Dataset acquisition or data collection	Weeks 1 to 3	Escalate or narrow scope if records are not confirmed by Week 3
Data cleaning, annotation, and preprocessing	Weeks 3 to 5	Depends on the received format and quality
System design	Weeks 3 to 5	Data schema, architecture, and dashboard flow
Algorithm or model development	Weeks 5 to 9	Build baselines before complex models
System implementation	Weeks 6 to 11	Dashboard developed in parallel
Model training and validation	Weeks 9 to 12	Rolling-origin evaluation
Software and user testing	Weeks 11 to 14	Requires partner or representative users
Performance evaluation	Weeks 12 to 14	Compare models and error by commodity
Revision and improvement	Weeks 14 to 16	Correct defects and refine explanations
Documentation	Weeks 4 to 17	Maintain methods and provenance as work proceeds
Deployment and final presentation	Weeks 17 to 18	Subject to the course calendar

Activities that may cause delays. The largest risks are data authorisation, poor data quality on arrival, and unavailable partner testers. These are managed by a Week 3 decision point: continue with the full title if BAPTC price and volume data are confirmed, otherwise adopt an adviser-approved narrower price-forecasting scope using accessible DA Bantay Presyo data.

Schedule finding. The 18-week plan is achievable, but it holds less slack than the other titles because the fallback still depends on assembling a usable secondary series rather than switching to something already in hand.

9. Legal and Ethical Feasibility

Requirement	Planned Control
Data Privacy Act	Apply data minimisation, purpose limitation, access control, secure storage, retention limits, and approved disposal. The study is designed around commodity-level aggregates and does



compliance	not collect trader or farmer identities.
Informed consent	Not expected for aggregate historical records. Consent is obtained if user testing collects identifiable responses.
Research ethics approval	Request an institutional ethics determination rather than assuming the project is exempt. Confirm the applicable review category before data collection, bulletin retrieval, or user testing.
Confidentiality and anonymisation	Use commodity aggregates, remove identifiers, restrict raw data, and report only approved summaries. If BAPTC data contains personal, confidential, or transaction-level fields, those fields must be excluded or anonymised before transfer and analysis.
Intellectual property and copyright	Cite sources, document model and software licences, and respect BAPTC and DA ownership and publication terms.
Licences	All proposed libraries carry permissive open-source licences compatible with academic use.
Data ownership and sharing restrictions	Use written authorisation or an agreement stating access, reuse, retention, publication, and deletion conditions.
Cybersecurity	Encrypt transfers, validate uploads, protect credentials, maintain backups, and log administrative activity.
AI fairness, transparency, accountability, bias	Report error by commodity and season; disclose uncertainty, missing data, thresholds, and known bias. A model that performs well on average may perform poorly on exactly the commodities where a glut matters most, so per-commodity error is reported rather than a single headline figure.
Possible misuse or harm	A forecast acted on as a certainty could cause real financial loss to a farmer. Forecasts are labelled as decision support, unsupported public claims are restricted, and the dashboard never presents a predicted price as a guaranteed one.

Legal and ethical finding. The aggregate-data design lowers privacy risk substantially, but written data-use conditions and an institutional ethics determination remain necessary before the full study proceeds.

10. Risk Assessment and Feasibility Conclusion

Identified Risk	Likelihood	Impact	Mitigation Strategy
Dataset is unavailable or insufficient	Medium	High	BAPTC has said it does not release raw files, so the request now asks for an aggregate extract. Hold a Week 3 decision gate and narrow to price forecasting on Bantay Presyo and weather data if no extract is confirmed.
Volume data withheld while price data is released	Medium	High	The glut-risk claim is dropped and the study narrows to price forecasting. This is a scope change, not a failure, and is decided at the Week 3 gate.
Technology is difficult to implement	Low	Medium	Use an open-source Python stack and a Streamlit prototype before adding integration complexity.
Algorithm produces poor results	Medium	High	Use naive and seasonal baselines, rolling-origin validation, feature review, and commodity-specific error analysis. A model that cannot beat seasonal-naive is reported as such.
Required respondents are unavailable	Medium	Medium	Recruit a small staff sample early and prepare remote or representative usability testing.
Privacy or ethical concerns arise	Low	High	Request only aggregates, secure raw data, obtain an ethics determination, and document authorisation.
Project exceeds the budget	Low	Medium	Use open-source tools, free tiers, existing devices, and a capped travel allowance. Travel to La Trinidad is the dominant cost and can be reduced by using the remote request channel.
Project cannot be completed on time	Medium	High	Time-box data access, prioritise a minimum viable dashboard, and trigger the narrower backup scope at Week 3.

Conclusion for this title: FEASIBLE WITH MODIFICATIONS.



Supporting findings. To conclude, the software, the algorithms, the operating environment, the cost, and the planned safeguards are all achievable for a student group, which means this title can be built with the skills and the equipment the group already has. The request reached the BAPTC General Manager and was acknowledged, but BAPTC has also said it does not release raw or source files, so the primary data source is still the single condition that decides this title. The title may proceed once the group confirms an exportable aggregate of historical BAPTC price and volume records, research-use permission, adequate coverage, and a realistic delivery date. Yet if those conditions are not met by Week 3, the scope should narrow to a price-only forecasting study using DA Bantay Presyo bulletins and PAGASA-BSU weather records, with the supply-glut claim withdrawn.

Provisional Feasibility Score for This Title

Scores use the activity's 1 to 5 scale, where 5 is highly feasible. The group's ranking across all five titles is carried in the final comparison table.

Criterion	Score	Basis
Technical and technology feasibility	4	Open-source stack is practical and familiar
Dataset and data feasibility	3	Request acknowledged, but BAPTC does not release raw files; an aggregate extract is still to be confirmed
Algorithm or model feasibility	4	Multiple established approaches with clear baselines available
Operational feasibility	4	A responsive institutional contact at BAPTC; the individual staff testers are named at the first meeting
Economic feasibility	4	Low-cost stack, but the highest travel cost of the five titles
Schedule feasibility	3	Feasible only with an early data decision gate
Legal and ethical feasibility	4	Aggregate design lowers risk; approvals are still required
Risk manageability	4	A defined fallback scope on public price and weather data makes this risk controllable
Total score	30/40	
Group's ranking	1	See the final comparison table

References

- Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 785-794.
- Department of Agriculture. (n.d.). BAPTC to spin into corporation to provide farmers, traders more services, bigger income. da.gov.ph
- Department of Agriculture Cordillera Administrative Region. (n.d.). Bantay Presyo price monitoring bulletins. car.da.gov.ph
- Freedom of Information Philippines. (n.d.). Program documents of the Benguet Agri-Pinoy Trading Center. foi.gov.ph
- Hyndman, R. J., & Athanasopoulos, G. (2021). Forecasting: Principles and practice (3rd ed.). OTexts. otexts.com/fpp3/
- OpenGov Asia. (n.d.). The Philippines: Digital trading to transform Benguet's agri-business sector. opengovasia.com
- Philippine Information Agency. (n.d.). DA eyes digital trading, profiling at Benguet trade center. pia.gov.ph



Supporting Review of Related Literature and Studies

Proposal Thesis Title

Highland Vegetable Price and Supply Glut Forecasting at the Benguet Agri-Pinoy Trading Center Using XGBoost

Review of Related Literature (RRL/s)

The RRL helps the group understand the chosen area, the problem being addressed, and the concepts or approaches relevant to the proposed solution.

Local Literatures:

1. **BAPTC to Spin into Corporation to Provide Farmers, Traders More Services, Bigger Income**

Description / Summary (Local Literature). The Department of Agriculture announced in 2021 that BAPTC, which it describes as the country's biggest vegetable trading facility and which began trading in 2015, would be converted into a corporation. The stated aim was to run the center more efficiently and to add services such as input trading, processing, and a savings and loan unit, so that farmers and traders earn a more reasonable return.

Relevance to the proposed study. It establishes BAPTC as the central trading point for highland vegetables and shows that the DA itself frames farmer income at BAPTC as a problem to be solved, which is the income the proposed glut warnings aim to protect.

Source: <https://www.da.gov.ph/baptc-to-spin-into-corporation-to-provide-farmers-traders-more-services-bigger-income/>

2. **DA Eyes Digital Trading, Profiling at Benguet Trade Center**

Description / Summary (Local Literature). The Philippine Information Agency reports that DA-CAR plans to introduce digital trading and profiling at BAPTC, including the use of artificial intelligence, to make the market more transparent and competitive. The first phase publishes the names of farms certified under Good Agricultural Practices.

Relevance to the proposed study. It shows that the agency that would use the proposed system is already moving BAPTC toward digital, data-driven tools, so a forecasting dashboard fits a direction DA-CAR has chosen rather than one the group is imposing.

Source: <https://pia.gov.ph/news/da-eyes-digital-trading-profiling-at-benguet-trade-center/>

3. **The Philippines: Digital Trading to Transform Benguet's Agri-Business Sector**

Description / Summary (Local Literature). OpenGov Asia describes the same DA-CAR programme as a phased modernisation of BAPTC that aims to give farmers, traders, and entrepreneurs better market tools, with transparency and competitiveness as the goals.

Relevance to the proposed study. It confirms from a second outlet that digital market information at BAPTC is a live government priority, and it supports the claim that farmers currently lack forward-looking market signals.

Source: <https://opengovasia.com/the-philippines-digital-trading-to-transform-benguets-agri-business-sector/>

4. **DA Bantay Presyo Price Monitoring**

Description / Summary (Local Literature). Bantay Presyo is the Department of Agriculture's price monitoring programme. Regional teams record prevailing prices of agricultural commodities, including highland vegetables, and publish daily and weekly price reports.



Relevance to the proposed study. It is the public, independently available price series the study uses to cross-check BAPTC data and as the fallback source for a price-only study if BAPTC volume data is not released.

Source: <https://www.da.gov.ph/price-monitoring/>

5. **Program Documents of the Benguet Agri-Pinoy Trading Center (BAPTC)**

Description / Summary (Local Literature). A request filed on the government's Freedom of Information portal for BAPTC's programme documents was approved, and the BAPTC Project Management Office released a BAPTC briefer while stating that it could not provide the conceptual framework.

Relevance to the proposed study. It shows what BAPTC documentation is publicly obtainable and how, which informs the group's own request and supports framing the request around aggregates rather than raw records.

Source: <https://www.foi.gov.ph/requests/program-documents-of-the-benguet-agri-pinoy-trading-center-baptc/>

Foreign Literatures:

6. **Forecasting: Principles and Practice (3rd ed.)**

Description / Summary (Foreign Literature). Hyndman and Athanasopoulos present the standard methods for time-series forecasting, including seasonal naive benchmarks, ARIMA and seasonal ARIMA, and time-series cross-validation in which the model is only ever tested on periods after the ones it was trained on.

Relevance to the proposed study. It is the basis for the study's rolling-origin validation and for requiring every model to beat naive and seasonal-naive forecasts before any result is reported.

Source: <https://otexts.com/fpp3/>

7. **XGBoost: A Scalable Tree Boosting System**

Description / Summary (Foreign Literature). Chen and Guestrin introduce XGBoost, a gradient-boosted decision tree system designed for speed and accuracy on structured data, with regularisation to limit overfitting and built-in handling of missing values.

Relevance to the proposed study. It is the algorithm named in the title. Its handling of missing values matters because BAPTC and bulletin series have gaps, and it supports both the price regression and the glut classification in one model family.

Source: <https://doi.org/10.1145/2939672.2939785>

8. **Analysis of Factors Affecting Vegetable Price Fluctuation: A Case Study of South Korea**

Description / Summary (Foreign Literature). Qiao, Kang, and Ahn separate the causes of Korean vegetable price swings into demand, domestic production, imports, and exports, and find that domestic production drove most of the recent fluctuation in garlic and onion prices.

Relevance to the proposed study. It supports using supply-side signals, such as arrivals and trading volume, as features, which is why the study treats BAPTC volume as essential for the glut-risk claim.

Source: <https://doi.org/10.3390/agriculture13030577>

9. **ARIMA Model Forecasting Analysis of the Prices of Multiple Vegetables under the Impact of the COVID-19**

Description / Summary (Foreign Literature). Mao and colleagues applied ARIMA to more than 51,000 vegetable price records collected during the COVID-19 pandemic and found that price behaviour during the disruption differed from normal seasons.

Relevance to the proposed study. It warns that a model trained on normal periods can fail during shocks, which is why the study reports error by season and period rather than one average figure.

Source: <https://doi.org/10.1371/journal.pone.0271594>



10. Machine Learning Outperforms Classical Forecasting on Horticultural Sales Predictions

Description / Summary (Foreign Literature). Haselbeck and colleagues compared nine machine learning models with three classical forecasting models on horticultural sales, using weather, holiday, and other features, and found machine learning ahead, with XGBoost best in 14 of 15 comparisons.

Relevance to the proposed study. It is direct support for choosing XGBoost for a horticultural market and for adding weather and calendar features.

Source: <https://doi.org/10.1016/j.mlwa.2021.100239>

Review of Related Studies (RRS/s)

The RRS focuses on existing research, systems, applications, or projects that address a problem similar to the one the group wants to solve.

Local Studies:

1. Hinterland Lettuce Selling Price Forecasting Model Using Recurrent Neural Networks

Description / Summary (Local Study). Lubag and colleagues tested ways of handling a broken, discontinuous BAPTC lettuce price series and optimised a bidirectional GRU model, which reached a test R-squared of 0.95 with an MAE of 8.97.

Relevance to the proposed study. It is the closest prior work on BAPTC prices and gives the study a benchmark. It forecasts one commodity's price only and does not classify gluts, which is the gap the proposed study fills.

Source: <https://doi.org/10.1109/HNICEM69498.2025.11569264>

2. The Analysis of Crop Production Data for Predicting Future Trends of Highland Vegetables in Benguet

Description / Summary (Local Study). Quitaleg and Dumlao forecast Benguet highland vegetable production with regression, support vector machines, decision trees, and random forest. Random forest performed best, and the authors note that complete farmer-level planting data would be needed for better results.

Relevance to the proposed study. It shows that Benguet production is forecastable and that supply-side data is the limiting factor, which supports the study's reliance on BAPTC volume as the supply signal.

3. From Price Stability to Food Security: What Are the Drivers of Price Volatilities in Philippine Vegetable Markets?

Description / Summary (Local Study). Hinlo and Lee decomposed the variance of Philippine onion and garlic prices from 2010 to 2024 and found that demand explained 50 to 90 percent of it and supply shocks 9 to 50 percent.

Relevance to the proposed study. It is Philippine evidence that price and supply both have to be modelled, supporting a system that forecasts price and flags supply gluts together.

Source: <https://doi.org/10.1186/s40066-025-00602-8>

Foreign Studies:

4. A Hierarchical RF-XGBoost Model for Short-Cycle Agricultural Product Sales Forecasting

Description / Summary (Foreign Study). Li and colleagues combined random forest and XGBoost with residual clustering to forecast 42 short-cycle products in six vegetable groups, reducing MAPE by 10 to 12 percent against either model alone.

Relevance to the proposed study. It shows XGBoost working on perishable vegetable data at the commodity level, which is the structure of the BAPTC data.

Source: <https://doi.org/10.3390/foods13182936>

5. **Research on Retail Pricing Strategy of Supermarket Fresh Products Based on XGBoost Model**

Description / Summary (Foreign Study). Zeng, Gao, and Hu used LSTM to forecast wholesale vegetable prices and XGBoost to estimate sales volume, reaching an R-squared of at least 0.93 on the price forecasts.

Relevance to the proposed study. It supports a two-output design in which one model forecasts price and another estimates volume, which is the design the glut alert needs.

6. **Exogenous Variable Driven Deep Learning Models for Improved Price Forecasting of TOP Crops in India**

Description / Summary (Foreign Study). Nayak and colleagues compared statistical, machine learning, and deep learning models for tomato, onion, and potato prices, and found that models using weather as an input outperformed those that did not.

Relevance to the proposed study. It is the reason the study adds PAGASA-BSU weather records as features.

Source: <https://doi.org/10.1038/s41598-024-68040-3>

7. **Machine Learning Techniques for Forecasting Agricultural Prices: A Case of Brinjal in Odisha, India**

Description / Summary (Foreign Study). Paul and colleagues forecast wholesale brinjal prices in 17 markets with GRNN, SVR, random forest, gradient boosting, and ARIMA. GRNN was best overall, while random forest was competitive in four markets.

Relevance to the proposed study. It shows that the best model differs by market, so XGBoost has to be benchmarked against alternatives rather than assumed best.

Source: <https://doi.org/10.1371/journal.pone.0270553>

8. **Forecasting Prices of Salad Vegetables Using Statistical and Artificial Intelligence Techniques**

Description / Summary (Foreign Study). Bulut compared CNN, LSTM, and random forest for 10-day-ahead vegetable price forecasts using Turkish price and trade-volume data, with CNN generally performing best.

Relevance to the proposed study. It shows that adding trade volume makes forecasts more useful in practice, which supports requesting BAPTC trading volume.

Source: <https://doi.org/10.1590/0103-8478cr20240444>

Additional Studies Also Consulted

Source	Type	Summary and relevance
Forecasting of Garlic Price Based on DA-RNN Using Attention Weight of Temporal Fusion Transformers	Foreign Study	Choi and colleagues forecast Korean garlic prices with an attention-based hybrid model (MAPE 4.25 percent) whose attention weights reveal which inputs mattered most. It shows that explaining which inputs drive a forecast is possible and useful, which the study achieves more simply with feature importance.
Short-Term Forecasting of Vegetable Prices Based on LSTM Model: Evidence from Beijing's Vegetable Data	Foreign Study	Zhang and colleagues applied LSTM to daily prices of six vegetable categories across seven Beijing markets from 2009 to 2023 (R-squared 0.958). It supports building models per commodity rather than one pooled model.
A Study on Short-Term Vegetable Price Prediction Based on the CNN-LSTM-Attention Model	Foreign Study	Bao and colleagues added uncertainty and extreme-weather variables to a CNN-LSTM-Attention model and improved short-term vegetable price prediction (MAPE 1.00 percent). It supports including weather shocks as inputs.
A Hybrid Model Integrating Variational Mode Decomposition and Intelligent Optimization for Vegetable Price Prediction	Foreign Study	Wang and colleagues combined signal decomposition, fruit-fly optimisation, and GRU for six vegetables from 2014 to 2024 and outperformed their baselines. It shows that more complex hybrids can score higher, but at a complexity the group judges too high for a first BAPTC system.



Vegetable Price Prediction Based on Optimized Neural Network Time Series Models	Foreign Study	Hou and colleagues found that optimised neural network models forecast stable commodities better than volatile ones. It suggests that volatile BAPTC commodities may need separate models or volatility features.
---	---------------	---

Research and System Gap

The reviewed work shows that vegetable prices are seasonal, nonlinear, and driven by both supply and demand, that weather and trading volume improve forecasts, and that XGBoost performs well on structured agricultural data. Most studies measure only forecast error and rarely connect the forecast to a decision a farmer or market manager has to make.

The gap. Few studies forecast price and supply glut together. The only BAPTC-specific price model found forecasts one commodity and does not flag gluts, and no study gives BAPTC or DA-CAR a commodity-level early warning of oversupply. The proposed system addresses this by forecasting price and volume per commodity and raising a glut-risk alert defined transparently, for example expected arrivals above the seasonal 75th percentile while forecast price is below the seasonal 25th percentile.

Refined Proposed Title and Solution

Highland Vegetable Price and Supply Glut Forecasting at the Benguet Agri-Pinoy Trading Center Using XGBoost (16 words). The title is kept. The review strengthened the solution rather than the wording: weather is added as a feature, every model is compared against seasonal-naïve, SARIMA, and random forest baselines under rolling-origin validation, and the glut threshold is stated explicitly.

References for the RRL and RRS

- Bao, W., Su, W., Zhao, X., et al. (2025). A study on short-term vegetable price prediction based on the CNN-LSTM-Attention model. *Discover Food*, 5, 176. <https://doi.org/10.1007/s44187-025-00426-2>
- Bulut, M. (2025). Forecasting prices of salad vegetables using statistical and artificial intelligence techniques. *Ciência Rural*, 55(11). <https://doi.org/10.1590/0103-8478cr20240444>
- Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785-794. <https://doi.org/10.1145/2939672.2939785>
- Choi, E., Park, Y., Choi, J., Choi, J., & Mesicek, L. (2023). Forecasting of garlic price based on DA-RNN using attention weight of temporal fusion transformers. *Mathematical Biosciences and Engineering*, 20(5), 9041-9061. <https://doi.org/10.3934/mbe.2023397>
- Department of Agriculture. (2021). BAPTC to spin into corporation to provide farmers, traders more services, bigger income. <https://www.da.gov.ph/baptc-to-spin-into-corporation-to-provide-farmers-traders-more-services-bigger-income/>
- Department of Agriculture. (n.d.). Price monitoring (Bantay Presyo). <https://www.da.gov.ph/price-monitoring/>
- Freedom of Information Philippines. (n.d.). Program documents of the Benguet Agri-Pinoy Trading Center (BAPTC). <https://www.foi.gov.ph/requests/program-documents-of-the-benguet-agri-pinoy-trading-center-baptc/>
- Haselbeck, F., Killinger, J., Menrad, K., Hannus, T., & Grimm, D. G. (2022). Machine learning outperforms classical forecasting on horticultural sales predictions. *Machine Learning with Applications*, 7, 100239. <https://doi.org/10.1016/j.mlwa.2021.100239>
- Hinlo, J., & Lee, Y. (2026). From price stability to food security: What are the drivers of price volatilities in Philippine vegetable markets? *Agriculture & Food Security*, 15, 28. <https://doi.org/10.1186/s40066-025-00602-8>



- Hou, Y., Sun, T., Cui, Y., et al. (2025). Vegetable price prediction based on optimized neural network time series models. *Smart Agriculture*, 7(5), 78-87. <https://doi.org/10.12133/j.smartag.SA202410037>
- Hyndman, R. J., & Athanasopoulos, G. (2021). *Forecasting: Principles and practice* (3rd ed.). OTexts. <https://otexts.com/fpp3/>
- Li, J., Lin, B., Wang, P., et al. (2024). A hierarchical RF-XGBoost model for short-cycle agricultural product sales forecasting. *Foods*, 13(18), 2936. <https://doi.org/10.3390/foods13182936>
- Lubag, M., Perocho, P. L., Co, R., Concepcion, R., Del-Amen, J., & Maniquiz-Redillas, M. (2025). Hinterland lettuce selling price forecasting model using recurrent neural networks. 2025 IEEE HNICEM. <https://doi.org/10.1109/HNICEM69498.2025.11569264>
- Mao, L., Huang, Y., Zhang, X., Li, S., & Huang, X. (2022). ARIMA model forecasting analysis of the prices of multiple vegetables under the impact of the COVID-19. *PLOS ONE*, 17(7), e0271594. <https://doi.org/10.1371/journal.pone.0271594>
- Nayak, G. H. H., Alam, M. W., Singh, K. N., et al. (2024). Exogenous variable driven deep learning models for improved price forecasting of TOP crops in India. *Scientific Reports*, 14, 17203. <https://doi.org/10.1038/s41598-024-68040-3>
- OpenGov Asia. (n.d.). The Philippines: Digital trading to transform Benguet's agri-business sector. <https://opengovasia.com/the-philippines-digital-trading-to-transform-benguets-agri-business-sector/>
- Paul, R. K., Yeasin, M., Kumar, P., et al. (2022). Machine learning techniques for forecasting agricultural prices: A case of brinjal in Odisha, India. *PLOS ONE*, 17(7), e0270553. <https://doi.org/10.1371/journal.pone.0270553>
- Philippine Information Agency. (n.d.). DA eyes digital trading, profiling at Benguet trade center. <https://pia.gov.ph/news/da-eyes-digital-trading-profiling-at-benguet-trade-center/>
- Qiao, Y., Kang, M., & Ahn, B.-I. (2023). Analysis of factors affecting vegetable price fluctuation: A case study of South Korea. *Agriculture*, 13(3), 577. <https://doi.org/10.3390/agriculture13030577>
- Quitaleg, A. R. M., & Dumlao, M. F. (2025). The analysis of crop production data for predicting future trends of highland vegetables in Benguet. *Journal of Interdisciplinary Perspectives*, 3(8).
- Wang, G., Xu, S., Chen, Z., & Li, Y. (2025). A hybrid model integrating variational mode decomposition and intelligent optimization for vegetable price prediction. *Agriculture*, 15(9), 919. <https://doi.org/10.3390/agriculture15090919>
- Zeng, L., Gao, Y., & Hu, H. (2024). Research on retail pricing strategy of supermarket fresh products based on XGBoost model. *Academic Journal of Computing & Information Science*, 7(10). <https://doi.org/10.25236/ajcis.2024.071012>
- Zhang, Q., Yang, W., Zhao, A., et al. (2024). Short-term forecasting of vegetable prices based on LSTM model: Evidence from Beijing's vegetable data. *PLOS ONE*, 19(7), e0304881. <https://doi.org/10.1371/journal.pone.0304881>



TARGET TITLE 2 OF 5

A Data Governance System for Verifiable CSR Impact Reporting Using Fellegi-Sunter Record Linkage

1. Target Thesis Title

A Data Governance System for Verifiable CSR Impact Reporting Using Fellegi-Sunter Record Linkage

Thesis specialization: Data Science

Primary technique: Data Governance and Ethics, from the approved Data Science technique list, supported by Data Mining and Data Visualization

Research domain: Beneficiary data quality and impact reporting for corporate social responsibility programmes

Partner: SM Foundation, Kabalikat sa Kabuhayan (KSK) farmers training programme

Proponents: Dawn Andrei Pamesa (Leader), Rogelio Porras, John Ashley Alba, Ralph Blanco

2. Project Overview

Background of the problem. SM Foundation's Kabalikat sa Kabuhayan programme has trained farmers in sustainable agriculture since 2007, in numbered batches of about fourteen weeks at training sites across Luzon, the Visayas, and Mindanao, and has passed its four hundredth batch. The Foundation reports the programme's reach as a count of farmer graduates, and that figure feeds corporate social responsibility and sustainability reporting. The totals published in different Foundation communications do not match: the programme page gives 28,550 graduates, while 2025 news coverage says over 30,000 and other communications say over 32,000.

Specific problem. The programme has no defensible, deduplicated count of the farmers it has trained. The total is built by adding up independent site and batch records over nearly twenty years without one shared identifier for each farmer, so a farmer who attends two batches, or whose name is spelled differently at two sites, can be counted twice, and nobody can say how often this happens. This is a measurement problem common to large multi-site programmes, not an error on the Foundation's part.

Target users and beneficiaries. The primary users are SM Foundation programme and reporting staff. SM Investments' sustainability reporting is a secondary beneficiary, and the farmers benefit indirectly from a programme that can show its reach accurately.

Proposed solution. A probabilistic record linkage system that reads the existing KSK training records, decides which entries describe the same farmer, and reports a deduplicated beneficiary count with a stated margin of error, delivered with a linkage quality report and a review dashboard.

Main objectives.

- Profile the KSK training records for completeness, field consistency, and duplication indicators.
- Implement blocking and comparison strategies suited to Philippine personal names, including phonetic encoding and approximate string comparison.
- Implement the Fellegi-Sunter model to classify record pairs as links, possible links, or non-links, with both error thresholds set explicitly.
- Evaluate linkage quality with precision and recall on a manually reviewed sample of record pairs.
- Produce a deduplicated beneficiary count with an uncertainty interval, and deliver the linkage layer as a reusable system.

Expected output. A data quality baseline of the KSK records, a deduplicated beneficiary spine, a linkage quality report, a review dashboard, and a disclosure-ready beneficiary figure with stated uncertainty.

Intended contribution. The first application of probabilistic record linkage to a Philippine corporate livelihood programme with no shared identifier, and the first measurement of how linkage methods perform on Philippine name conventions.

3. Technical and Technology Feasibility

Area	Proposed Choice	Feasibility and Alternative
Programming languages	Python; SQL; R only for cross-checking	Python covers profiling, linkage, evaluation, and the dashboard in one stack.
Frameworks and libraries	pandas, recordlinkage or Splink, jellyfish for phonetic codes, R fastLink for cross-checking	All are open source and implement Fellegi-Sunter directly, so the group configures and evaluates the model rather than writing it from scratch.
Database and storage	SQLite, with CSV and XLSX import	SQLite needs no server; the restricted extract stays on encrypted storage.
APIs and third-party services	None	Nothing on the critical path depends on an outside service.
Platform	Streamlit dashboard	A single-user dashboard is enough for Foundation staff; FastAPI is an option if integration is later wanted.
Hardware and equipment	Development laptops	Blocking keeps the comparisons tractable; no GPU is needed.
Architecture and integration	Restricted import with a prohibited-field check, then profiling, blocking, comparison, Fellegi-Sunter classification, clerical review, the deduplicated spine, and the quality report	The system reads exports and never writes into the Foundation's own systems.
Security and maintenance	Encrypted storage, access limited to the group, hashed or phonetic name keys only, audit log of review decisions	The group never holds plain-text names.
Technical expertise	Python and pandas are in hand. Record linkage evaluation is the least familiar area.	The evaluation measures are well documented by Christen and by Hand and Christen.
Limitations and alternatives	Linkage quality depends on the fields the Foundation keeps.	If names cannot be hashed on the Foundation's side, linkage runs on the Foundation's computers and only results come back.

Technical finding. The stack is open source and the named algorithm has mature reference implementations. The technical constraint is the content of the records, not the software.

4. Dataset and Data Feasibility

Requirement	Current Assessment
Dataset name and description	SM Foundation KSK farmer training records: an opaque record ID, hashed or phonetic name keys, batch number, training site, province, training dates, and crop or training track.
Source, owner, and provider	SM Foundation. Dawn raised the partnership in person at a Foundation volunteering activity on September 21, 2026 and was invited to email the proposal. The proposal letter was emailed to education@sm-foundation.org on September 23, 2026, and a reply is pending.



Public link	No public dataset. The programme page and news coverage are public evidence of the problem, not the data.
Data type and file format	Tabular records; CSV, XLSX, or a database export.
Estimated size	Over 30,000 graduate records across more than 400 batches by the Foundation's own public figures, above the 10,000-record floor.
Features and attributes	Name comparison keys, batch, site, province, training dates, crop or track.
Labels and target variables	A manually reviewed sample of 300 to 500 record pairs labelled as the same farmer or not, requested from the Foundation or built by the group under supervision.
Accessibility and collection	Through the invited proposal. The letter asks only for agreement in principle first; the data request follows separately.
Permission and licensing	Written data-use terms covering access, retention, deletion, and the group's right to publish the method and aggregate findings.
Quality, completeness, and bias	Early batches may be paper only; name spellings vary; married-name changes break surname agreement.
Preprocessing and storage	Standardise sites, provinces, and dates; phonetic-encode names on the Foundation's side; store on encrypted media.
Training, validation, testing	The labelled pair sample is split for threshold tuning and a held-out evaluation.
Alternative data plan	If only part of the history is digital, link the digitised period and state the coverage. If records are paper only, the title does not proceed on this partner.

Dataset finding. Feasibility is conditional. The channel is warm and invited rather than cold, but the record format, the digital coverage, and the ground-truth sample are all unconfirmed.

5. Algorithm or Model Feasibility

Evaluation Area	Approach 1	Approach 2	Approach 3
Name of algorithm or approach	Deterministic exact matching	Fuzzy string threshold (Jaro-Winkler)	Fellegi-Sunter probabilistic record linkage
Purpose in the proposed project	Naive baseline	Common practical baseline	Primary linkage model
Dataset requirements	Clean identical fields	One or two name fields	Several comparison fields; a labelled sample for evaluation
Expected performance	High precision, very low recall on variant names	Better recall, but the cutoff has no error meaning	Chosen error rates on both sides
Computational requirements	Low	Low	Low to moderate, with blocking
Advantages	Simple and transparent	Handles spelling variants	Decision-theoretic; sends only uncertain pairs to review
Limitations	Misses every variant	Arbitrary cutoff	Parameter estimation needs care with missing fields
Evaluation metrics	Precision, recall	Precision, recall	Precision and recall across thresholds; reduction ratio and pairs completeness for blocking
Implementation complexity	Low	Low	Moderate

Recommended approach. Fellegi-Sunter is the primary model, because a disclosure figure needs error rates that were chosen rather than a similarity cutoff nobody can defend. Exact matching and a Jaro-Winkler threshold are kept as baselines. Missing fields follow the data-adaptive approach of Li, Xu, and Grannis (2022), and precision and recall are reported separately rather than as a single F-measure, following Hand and Christen (2018).

6. Operational Feasibility

Area	Assessment
Target users and stakeholders	SM Foundation programme and reporting staff; SM Investments' sustainability reporting as a secondary user.
Intended operating environment	Office computers with a browser; the dashboard runs locally.
Availability of participants	A Foundation contact who knows the programme is needed to explain fields and review a sample. The Foundation invited the proposal, but no named participant has yet agreed.
User needs and benefits	A beneficiary figure the Foundation can state and defend, with a documented method.
Devices and facilities	Standard office computers. No procurement.
Usability and accessibility	Plain-language quality report; review queue tested with Foundation staff.
Training and support	A short user guide for rerunning the linkage on new batches.
Maintenance	New batches are linked incrementally against the existing spine.
Adoption challenges	The Foundation may read the result as a correction of its published figures. The study frames it as a measurement improvement, not an error.

Operational finding. The Foundation is both the data owner and the user, which keeps the operational path short once it agrees.

7. Economic Feasibility

Expense	Estimated Cost	Purpose	Possible Alternative
Software licences	PHP 0	Open-source linkage and dashboard tools	None needed
APIs and subscriptions	PHP 0	None	None
Cloud hosting and storage	PHP 0 to 1,000	Encrypted storage and optional hosting	Local deployment
Dataset acquisition or annotation	PHP 0	Review of labelled pairs by the group	Foundation-supplied sample
Hardware or equipment	PHP 0	Existing laptops	None
Transportation and data collection	PHP 500 to 1,500	Meetings at the Foundation in Metro Manila	Online meetings
Testing and deployment	PHP 500	User review session	Remote session
Contingency expenses	PHP 1,000	Unexpected costs	Reduce optional hosting
Total estimated cost	PHP 2,000 to 4,000	Working budget range	Lower bound uses free tiers

Economic finding. Low cost. The Foundation is in Metro Manila, so travel is minor.

8. Schedule Feasibility



Activity	Estimated Duration	Dependency or Control
Requirements gathering	Weeks 1 to 2	Coverage questions to the Foundation; feasibility gate
Literature and related-study review	Weeks 1 to 4	Continues through method design
Dataset acquisition	Weeks 2 to 4	Decision point in Week 4
Data profiling and preprocessing	Weeks 4 to 6	Depends on the received format
System design	Weeks 3 to 5	Pipeline and dashboard flow
Model development	Weeks 5 to 9	Blocking, comparison, Fellegi-Sunter fit
System implementation	Weeks 6 to 11	Review queue and dashboard in parallel
Evaluation	Weeks 9 to 13	Precision and recall on the labelled sample
Software and user testing	Weeks 11 to 14	Foundation staff review
Revision and improvement	Weeks 14 to 16	Threshold and report refinement
Documentation	Weeks 4 to 17	Maintained throughout
Deployment and final presentation	Weeks 17 to 18	Handover of the linkage layer

Activities that may cause delays. The Foundation's reply time and the format of its records. Managed by a Week 4 decision point: proceed if digital records and a hashing arrangement are confirmed; otherwise the group moves to its backup title.

Schedule finding. Achievable in 18 weeks if the Foundation answers early; the waiting time is the main schedule risk.

9. Legal and Ethical Feasibility

Requirement	Planned Control
Data Privacy Act compliance	Data minimisation: hashed or phonetic name keys only, municipality-level geography, no contact details.
Informed consent	Not needed for de-identified records; obtained for any staff usability session.
Research ethics approval	Exempt classification targeted, holding only if the field exclusions hold.
Confidentiality and anonymisation	4Ps status is means-tested and therefore sensitive personal information under RA 10173; it is excluded even if offered, as are income, household, health, disability, and PhilSys numbers.
Intellectual property and copyright	The linkage layer is handed to the Foundation at no cost; the group keeps the right to publish the method and aggregate findings.
Data ownership and sharing restrictions	Written data-use terms before any transfer; deletion at the end of the study.
Cybersecurity	Encrypted storage, restricted access, audit log.
AI fairness, transparency, accountability, bias	Report linkage error by province and name pattern so that no group is systematically over- or under-counted.
Possible misuse or harm	Batch plus site plus exact date can single out a person; date windows are used where linkage quality allows.

Legal and ethical finding. Exempt review is achievable because the group never holds identifiable records, provided the field exclusions are stated in writing and kept.



10. Risk Assessment and Feasibility Conclusion

Identified Risk	Likelihood	Impact	Mitigation Strategy
Records are paper for the early years	High	Medium	Link the digital period and state the coverage
No ground-truth sample available	Medium	High	Negotiate the labelled sample as part of the request, or build it under supervision
4Ps status arrives in the extract unasked	Medium	High	Exclusion stated in writing; if it arrives, it is not loaded and the adviser is told
Foundation reads the pitch as an accusation	Medium	High	Framed as a measurement improvement, not an error
Foundation reply is slow	Medium	Medium	Week 4 decision point
Project cannot be completed on time	Low	High	Mature libraries; minimum viable dashboard first

Conclusion for this title: FEASIBLE WITH MODIFICATIONS.

Supporting findings. To conclude, the method, the tools, and the cost are all within reach, and the problem is visible in the Foundation's own published figures. The partner channel is warm and the Foundation invited the proposal. However, whether the records are digital, whether names can be hashed on the Foundation's side, and whether a reviewed sample can be built are all unconfirmed, and those decide the title.

Provisional Feasibility Score for This Title

Scores use the activity's 1 to 5 scale, where 5 is highly feasible. The group's ranking across all five titles is carried in the final comparison table.

Criterion	Score	Basis
Technical and technology feasibility	4	Mature open-source linkage libraries
Dataset and data feasibility	3	Warm invited channel, but record format and ground truth unconfirmed
Algorithm or model feasibility	5	Named algorithm with a reference implementation
Operational feasibility	4	The Foundation is the data owner and the user
Economic feasibility	5	No paid tools; partner is in Metro Manila
Schedule feasibility	3	Depends on the Foundation's reply time
Legal and ethical feasibility	4	Exempt only if the field exclusions hold
Risk manageability	4	Clear go and no-go conditions
Total score	32/40	
Group's ranking	4	See the final comparison table

Supporting Review of Related Literature and Studies

Proposal Thesis Title

A Data Governance System for Verifiable CSR Impact Reporting Using Fellegi-Sunter Record Linkage



Review of Related Literature (RRL/s)

The RRL helps the group understand the chosen area, the problem being addressed, and the concepts or approaches relevant to the proposed solution.

Local Literatures:

1. Republic Act No. 11055, the Philippine Identification System Act

Description / Summary (Local Literature). The law established the Philippine Identification System, a single national identification system for citizens and resident aliens with a unique PhilSys Number for each registered person. It recognises that fragmented identification across many agencies creates inefficiency, and it includes provisions protecting registered information from unauthorised disclosure.

Relevance to the proposed study. It shows that the lack of a common identifier across records is a recognised Philippine problem. It also explains why the proposed study will not request PhilSys numbers, and will instead build the link from the fields the Foundation already keeps.

Source: https://lawphil.net/statutes/repacts/ra2018/ra_11055_2018.html

Foreign Literatures:

2. A Theory for Record Linkage

Description / Summary (Foreign Literature). Fellegi and Sunter (1969) gave record linkage its formal decision-theoretic foundation. Every pair of records is compared field by field, and the pattern of agreements is used to place the pair into one of three groups: a link, a possible link that needs a human to decide, or a non-link. They showed that the best rule is based on a ratio of how likely that agreement pattern is among true matches versus non-matches, and that this rule keeps the number of pairs sent for manual review as small as possible for the two error rates the user chooses.

Relevance to the proposed study. It is the algorithm named in the proposed title, and it is what allows the system to attach chosen error rates to the beneficiary count instead of relying on an arbitrary similarity cutoff.

Source: <https://doi.org/10.1080/01621459.1969.10501049>

3. Quality and Complexity Measures for Data Linkage and Deduplication

Description / Summary (Foreign Literature). Christen and Goiser (2007) examined how the quality and the cost of linkage and deduplication should be measured. They discussed measures for blocking, such as how much the number of candidate pairs is reduced and how many true matches survive that reduction, and they warned that some common classification measures become misleading because non-matching pairs vastly outnumber matching ones.

Relevance to the proposed study. It supplies the evaluation measures for the blocking and classification modules, so the system's results can be reported in the terms used by the record linkage field.

Source: https://doi.org/10.1007/978-3-540-44918-8_6

4. (Almost) All of Entity Resolution

Description / Summary (Foreign Literature). Binette and Steorts (2022) reviewed entity resolution, the task of finding which records refer to the same real-world entity when no shared unique identifier exists. The review walks through the stages of a typical pipeline, from blocking and comparison to classification and evaluation, and compares probabilistic, machine learning and Bayesian approaches.

Relevance to the proposed study. It frames the proposed system within the wider entity resolution field and supports the design of the system as a pipeline of separate blocking, comparison and classification modules.

Source: <https://doi.org/10.1126/sciadv.abi8021>

5. Challenges in Administrative Data Linkage for Research

Description / Summary (Foreign Literature). Harron et al. (2017) discussed the problems that arise when administrative records are linked for research. A central concern is bias from linkage errors, where records that

belong together are missed or records that do not belong together are joined, and the authors argue for transparent reporting of linkage quality alongside the results.

Relevance to the proposed study. It supports the system's requirement to report its beneficiary count with a stated uncertainty and a linkage quality report, instead of presenting a single number as exact.

Source: <https://doi.org/10.1177/2053951717745678>

Review of Related Studies (RRS/s)

The RRS focuses on existing research, systems, applications, or projects that address a problem similar to the one the group wants to solve.

Local Studies:

1. Enhancing CRVS Data Quality and Inclusivity through Integration with PhilSys

Description / Summary (Local Study). Du (2026), writing from the Philippine Statistics Authority, assessed early efforts to integrate civil registration with the Philippine Identification System in selected local government units. The paper identifies duplicate records as a persistent data quality problem and describes how PhilSys numbers and authentication can support record verification and reduce duplication. It found that structured verification, coordination between agencies and capacity-building were the key enablers.

Relevance to the proposed study. It is Philippine evidence that duplicate records are a recognised national data quality problem. The proposed study addresses the same problem for a programme where no shared identifier exists.

Source: https://getinthepicture.org/system/files/sites/default/files/B1_Session%203.2_102_Du_Paper_0.pdf

2. Development and Implementation of an Interoperable Local Health Information Exchange Platform for the Attainment of Universal Health Coverage in the Philippines

Description / Summary (Local Study). Zuniga, Carpio and Tayam (2026) described SMILHIS, an open-source health information exchange deployed in three Philippine local government units using the HL7 FHIR standard. It includes a client registry that serves as a central identity record for patients, and the authors propose PhilSys as the primary registry so that patient records can be deduplicated across facilities.

Relevance to the proposed study. It shows a Philippine system that needed a single identity record to stop the same person being counted in several places. The proposed study builds that kind of record for a programme through probabilistic linkage.

Source: <https://doi.org/10.3389/fdgth.2025.1650273>

Foreign Studies:

3. Using a Probabilistic Model to Assist Merging of Large-Scale Administrative Records

Description / Summary (Foreign Study). Enamorado, Fifield and Imai (2019) developed fastLink, a fast and scalable implementation of the Fellegi-Sunter model. It handles millions of records while accounting for missing data and measurement error, and it carries uncertainty about the merge into later analysis. They tested it through simulations and applied it to campaign contribution records, survey data and nationwide voter files.

Relevance to the proposed study. It shows the Fellegi-Sunter model working at a scale larger than the KSK records and gives the study an open-source reference implementation to compare against.

Source: <https://doi.org/10.1017/S0003055418000783>

4. The Data-Adaptive Fellegi-Sunter Model for Probabilistic Record Linkage

Description / Summary (Foreign Study). Li, Xu and Grannis (2022) tested ways of handling missing fields in the Fellegi-Sunter model. Treating missing values as missing at random maintained or improved F1-scores in all four of their use cases, and combining that assumption with data-driven selection of matching fields gave the best overall performance. They noted this is especially useful in privacy-preserving linkage.



Relevance to the proposed study. Programme records gathered across many sites and years will have gaps, and the study will link on hashed keys. This gives the study a tested way to handle missing fields instead of discarding those record pairs.

Source: <https://doi.org/10.2196/33775>

5. **A Note on Using the F-Measure for Evaluating Record Linkage Algorithms**

Description / Summary (Foreign Study). Hand and Christen (2018) showed that the F-measure is a poor basis for comparing record linkage methods. The relative weight it places on precision and recall depends on the number of pairs each method predicts as matches, so two methods end up being judged on different scales.

Relevance to the proposed study. It is the reason the proposed study will report precision and recall separately, and show how they trade off as the thresholds move, instead of relying on a single F-score.

Source: <https://doi.org/10.1007/s11222-017-9746-6>

Research and System Gap

Probabilistic record linkage is a mature and well-evaluated method. Fellegi-Sunter gives it a decision-theoretic basis, fastLink shows it working on millions of records, and later work shows how to handle missing fields and why linkage quality must be reported alongside the results. The Philippine sources show that duplicate records are a recognised national data quality problem, and that the government's answer has been a shared identifier such as PhilSys.

The gap. No published study applies probabilistic linkage to a Philippine corporate or NGO livelihood programme where no shared identifier exists and none can be added retroactively. No study measures linkage performance on Philippine name conventions, and none turns a programme's beneficiary count from a sum of site records into a figure with a stated error.

Refined Proposed Title and Solution

A Data Governance System for Verifiable CSR Impact Reporting Using Fellegi-Sunter Record Linkage (13 words). The title is kept, because the review confirmed that the named algorithm fits the problem.

References for the RRL and RRS

- Binette, O., & Steorts, R. C. (2022). (Almost) all of entity resolution. *Science Advances*, 8(12), eabi8021. <https://doi.org/10.1126/sciadv.abi8021>
- Christen, P., & Goiser, K. (2007). Quality and complexity measures for data linkage and deduplication. In *Quality measures in data mining* (pp. 127-151). Springer. https://doi.org/10.1007/978-3-540-44918-8_6
- Du, A. S. (2026). Enhancing CRVS data quality and inclusivity through integration with PhilSys. Paper presented at the 2026 CRVS Research Forum, Bangkok, Thailand. https://getinthepicture.org/system/files/sites/default/files/B1_Session%203.2_102_Du_Paper_0.pdf
- Enamorado, T., Fifield, B., & Imai, K. (2019). Using a probabilistic model to assist merging of large-scale administrative records. *American Political Science Review*, 113(2), 353-371. <https://doi.org/10.1017/S0003055418000783>
- Fellegi, I. P., & Sunter, A. B. (1969). A theory for record linkage. *Journal of the American Statistical Association*, 64(328), 1183-1210. <https://doi.org/10.1080/01621459.1969.10501049>
- Hand, D., & Christen, P. (2018). A note on using the F-measure for evaluating record linkage algorithms. *Statistics and Computing*, 28(3), 539-547. <https://doi.org/10.1007/s11222-017-9746-6>
- Harron, K., Dibben, C., Boyd, J., Hjern, A., Azimae, M., Barreto, M. L., & Goldstein, H. (2017). Challenges in administrative data linkage for research. *Big Data & Society*, 4(2). <https://doi.org/10.1177/2053951717745678>



- Li, X., Xu, H., & Grannis, S. (2022). The data-adaptive Fellegi-Sunter model for probabilistic record linkage. *Journal of Medical Internet Research*, 24(9), e33775. <https://doi.org/10.2196/33775>
- Republic Act No. 11055. (2018). Philippine Identification System Act. https://lawphil.net/statutes/repacts/ra2018/ra_11055_2018.html
- Zuniga, P., Carpio, J., Tayam, Q., Garcia, P., Lacsina, A., Calderon, J., et al. (2026). Development and implementation of an interoperable local health information exchange platform for the attainment of universal health coverage in the Philippines. *Frontiers in Digital Health*, 7. <https://doi.org/10.3389/fdgth.2025.1650273>



TARGET TITLE 3 OF 5

A Data Governance System for Privacy Release Readiness of Quezon City Open Data Using Mondrian k-Anonymity

1. Target Thesis Title

A Data Governance System for Privacy Release Readiness of Quezon City Open Data Using Mondrian k-Anonymity

Thesis specialization: Data Science

Primary technique: Data Governance and Ethics, from the approved Data Science technique list

Research domain: Statistical disclosure control, open government data, and privacy-preserving data publishing

Proponents: Dawn Andrei Pamesa (Leader), Rogelio Porras, John Ashley Alba, Ralph Blanco

2. Project Overview

Background of the problem. Quezon City publishes an Open Data Dashboard through its Information Technology Development Department together with source departments, with listed dashboards containing data from 2019 onward. Current programmes include Helpline 122, QCitizen ID, the PWD community, maintenance medicine, Pangkabuhayang QC, and scholarships, and the city has stated its intent to publish more. The official ITDD page lists a service called Control of Data Services in Conformity with the Data Privacy Act, under which an approved letter is forwarded to the Database Management Division for extraction, which is direct evidence that ITDD is the correct office to approach.

Specific problem. Every release has to answer two questions at once: whether the source data is fit for public use, and how much detail can be published without making a person identifiable. Removing names does not settle the second question. A rare category combined with a small geography and a narrow date can single out a person even when no direct identifier remains. At the same time, aggregating everything to be safe can make a dataset too coarse to answer any useful question. At present there is no documented procedure for comparing candidate publication granularities in terms of both residual disclosure risk and analytical utility, and no auditable record of why a given date, geography, category, or suppression level was chosen.

Target users and beneficiaries. The direct users are ITDD staff and source-department data owners who decide what a release should look like. The beneficiaries are the residents whose records appear in those datasets, and the researchers, journalists, and civic groups who use QC open data and currently have no documented statement of what a published dataset can and cannot support.

Proposed solution. A web-based governance system that profiles an ITDD-approved de-identified extract, classifies every field as a direct identifier, quasi-identifier, sensitive attribute, or ordinary attribute, generates candidate release profiles by generalising dates, geography, and categories, measures residual disclosure risk and information loss for each profile using Mondrian k-anonymity, and produces an auditable release-readiness report recommending publication settings.

Main objectives.

- Define a release-readiness metadata profile for a selected QC Open Data source.
- Measure completeness, validity, consistency, uniqueness, and timeliness using versioned rules.
- Classify direct identifiers, quasi-identifiers, sensitive attributes, and non-identifying fields from the approved schema.
- Measure equivalence-class and small-cell risk under candidate publication granularities.
- Apply controlled generalisation and suppression without altering the restricted source copy.

- Quantify information loss and aggregate-query error produced by each strategy.
- Compare a fixed suppression baseline against Mondrian k-anonymity on the same risk threshold.
- Produce an auditable release-readiness report documenting assumptions and unresolved risks.

Expected output. A field governance classification for the selected dataset, a measured risk and utility comparison across candidate release profiles, a working release-readiness system with an aggregate dashboard, an exportable report, and a reusable configuration format adaptable to other QC datasets.

Intended contribution. An evaluated and reusable procedure for measuring the quality, risk, and utility trade-off of a specific Philippine local government open-data release. National Privacy Commission guidance establishes that de-identification is not automatically anonymisation, but no published work applies that standard as a measurable, auditable workflow to a Philippine city release.

3. Technical and Technology Feasibility

Area	Proposed Choice	Feasibility and Alternative
Programming languages	Python 3.11 for profiling, risk measurement, generalisation, and dashboard logic; SQL for staging queries; minor HTML and CSS for styling	Standard undergraduate skills already used by the group.
Frameworks and libraries	pandas and NumPy for profiling and equivalence-class computation; a custom Mondrian implementation, with ARX or pycanon as cross-check references; scikit-learn for encoding utilities; Plotly for risk and utility charts; Streamlit for the dashboard	Mondrian is roughly 200 lines of recursive partitioning code and is well documented in the literature, so a custom implementation is realistic and also makes the mechanism defensible at the defence.
Database and storage	SQLite for the immutable staging copy and release-profile versions; PostgreSQL if shared deployment is later required; CSV or database export accepted on import	No server administration needed during the study.
APIs and third-party services	None on the critical path. The QC Open Data Dashboard is inspected manually through the browser to record what the public surface exposes.	Nothing behind a login is scraped or automated, which is both a policy requirement and a removal of technical risk.
Platform	Browser-based application run locally. A free-tier host is used only for a synthetic or already-public demonstration dataset.	The restricted record-level extract is never hosted externally under any circumstance.
Hardware and equipment	Existing group laptops with 8 GB RAM. No sensors, no GPU.	Mondrian partitioning on tens of thousands of rows completes in seconds to minutes on a CPU.
Architecture and integration	ITDD-approved de-identified source copy, then prohibited-field and provenance gate, then quality profiler, then field governance classifier, then candidate release-profile generator, then equivalence-class and small-cell risk engine, then generalisation and suppression engine, then utility evaluator, then release-readiness report and dashboard	File-based import only. No connection to any city production system, which is an ethics requirement as much as an engineering choice.
Security and maintenance	Restricted external storage path outside Git and outside the thesis vault; checksum, schema version, and provenance recorded on import; prohibited-field scan before any analytical module loads data; no linkage attempt against any external source; audit logging	The no-linkage rule is enforced by design, since the system holds no external dataset to link against.
Technical expertise	Python, pandas, SQL, and Streamlit are all in hand. Mondrian and equivalence-class risk	ARX or pycanon serve as reference implementations to validate the custom Mondrian



	measurement are new to the group but are well specified in published pseudocode.	output against, so a subtle implementation error can be caught rather than defended.
Limitations and alternatives	Generalisation hierarchies cannot be finalised until the approved schema and data dictionary are seen.	Hierarchies are drafted against the public dashboard's visible fields, and the pipeline is developed and validated on a public benchmark microdata set such as the UCI Adult dataset before the real extract arrives.

Technical finding. The stack is open source and familiar, and the only genuinely new component, Mondrian, has published pseudocode and reference implementations available for cross-checking. The critical path holds no external API and no specialised hardware. The technical risk is low; the constraint is data access, treated in section 4.

4. Dataset and Data Feasibility

Requirement	Current Assessment
Dataset name and description	One ITDD-approved de-identified record-level dataset from the QC Open Data programme, most likely Helpline 122 service-request records, requested after direct-identifier removal but before dashboard aggregation, together with its data dictionary, category definitions, refresh schedule, and known quality notes.
Source, owner, and provider	Quezon City Information Technology Development Department, with the originating source department as a probable co-approver.
Public link	The QC Open Data Dashboard is publicly viewable at quezoncity.gov.ph/program/qc-open-data-dashboard/ , and the city publishes a documented Request Information service. The dashboards present aggregated views; a record-level extract is not publicly downloadable and must be requested.
Data type and file format	Expected CSV or a database export. The dashboard surface must be inspected first to record which fields, filters, periods, and aggregation levels are already public, since that inspection needs nobody's permission and can be done in Week 1.
Estimated size	The study requires at least 10,000 usable rows. Helpline 122 is a citywide complaint and service channel with dashboards covering 2019 onward, so the volume is plausible, and the exact record count will be confirmed when ITDD replies to the request already sent.
Features and attributes	Request or transaction date, service or concern category and sub-category, barangay or district, status, resolution time, and operational metadata. Narrative text, exact locations, account identifiers, and direct identifiers are excluded at source and must not be received.
Labels and target variables	Not a prediction task, so no outcome label is required. The measured quantities are equivalence-class size distribution, minimum k, proportion of records in small cells, information loss, and aggregate-query error under each candidate release profile.
Accessibility and collection	Four steps. First, verify the public surface through the browser and record what is already exposed. Second, review the exact field list, ethics route, and letter wording with the adviser. Third, submit a formal request through QC's Request Information service addressed to ITDD. Fourth, receive under agreed terms only after ethics clearance. A complete letter draft already exists in the group's working files.
Permission and licensing	The approved scope, publication conditions, and any data-sharing requirements are set by ITDD and the source department. The group does not publish the approved record-level extract under any circumstance.
Quality, completeness, and bias	Expected issues include missing or inconsistent categories, changing status codes across years, unknown refresh delays, schema changes, and reporting bias toward barangays and populations that use the channel more. That reporting bias is reported as a limitation and is not corrected for, since correcting it would require assumptions the data cannot support.
Preprocessing and storage	Immutable restricted copy with checksum and provenance; prohibited-field scan before any analytical module runs; all work on versioned derivatives; no attempt to link the research copy to any public dataset or to recover identities.
Training, validation,	Not a supervised split. Evaluation compares release profiles on the same dataset across multiple



testing	values of k , reporting risk and utility for each. The pipeline is first validated on a public benchmark microdata set with published k -anonymity results, so correctness is established before the real extract is touched.
Alternative data plan	Three fallbacks in order. First, a different QC dataset if the first choice is refused, since the method is dataset-agnostic. Second, a Philippine Statistics Authority anonymised microdata product, which is a real and obtainable microdata source. Third, a published benchmark microdata set such as UCI Adult, which preserves the full method and every measurement but loses the local government finding and the practical contribution to the city.

Dataset finding. Stronger than most of the titles on this criterion, for two reasons that are both verifiable rather than hoped for: Quezon City operates a documented public request channel with a named responsible department, and a substantial part of the Week 1 work, inspecting the public dashboard surface, requires nobody's approval. The formal letter and email to ITDD were sent on September 23, 2026, following an earlier request through an identified contact in the city, so what is outstanding is the reply and the shape of the record-level extract, and the response time is the part that stays outside the group's control.

5. Algorithm or Model Feasibility

Evaluation Area	Approach 1	Approach 2	Approach 3
Name of algorithm or approach	Fixed small-cell suppression threshold, the current common practice baseline	Mondrian multidimensional k -anonymity through greedy recursive partitioning	Differential privacy using the Laplace mechanism on aggregate queries
Purpose in the proposed project	Establish the baseline that the city's present approach approximates, so any improvement is measured against real practice rather than a straw man	Generate release profiles that reach a declared k with less information loss than the baseline	Provide a formal-guarantee comparison point for aggregate statistics
Dataset requirements	Any tabular dataset with categorical fields	Identified quasi-identifiers and a generalisation hierarchy defined per attribute	A defined query set and a privacy budget; does not release record-level data at all
Expected performance	Meets the threshold but over-suppresses, discarding detail that a smarter partition would have kept	Expected to reach the same k with measurably lower information loss, which is the study's central hypothesis	Strong formal guarantee, but not comparable on a record-level release task
Computational requirements	Trivial	Low to moderate; recursive partitioning completes in seconds to minutes at this scale	Low for the noise mechanism, but budget accounting adds design overhead
Advantages	Simple, explainable, and already understood by data owners	Multidimensional partitioning preserves more utility than fixed global rules; the mechanism is inspectable and can be shown to a panel	Provides a mathematical guarantee that k -anonymity cannot
Limitations	Blunt and lossy; cannot adapt to which combinations are actually rare	k -anonymity is vulnerable to homogeneity and background-knowledge attacks, and is not a legal guarantee of anonymity	Answers a different question, since it supports aggregate queries rather than a record-level release, so it is a reference point and not a substitute
Evaluation metrics	Minimum k achieved, suppression rate, information loss	Minimum k , equivalence-class size distribution, discernibility metric, normalised certainty penalty, and aggregate-query error against the ungeneralised copy	Epsilon spent against measured query error



Implementation complexity	Very low	Moderate; roughly 200 lines with published pseudocode and reference implementations available for cross-checking	Moderate to high, mainly in budget accounting
---------------------------	----------	--	---

Recommended approach. Mondrian k-anonymity is the primary algorithm, because the research question is specifically whether a declared risk threshold can be reached with less information loss than blanket aggregation, and Mondrian is the established method that makes that comparison measurable on a record-level release. Fixed small-cell suppression is retained as the mandatory baseline, since a result is only meaningful against what data owners actually do today. Differential privacy is included in the comparison as a discussion reference and is explicitly out of scope for implementation, because it answers a different release question. k is treated as a configurable study parameter reported across multiple plausible values, never as a legal guarantee, and the final policy used in any conclusion is approved by the adviser or the data owner. The pipeline is validated against a public benchmark with published results before the city extract is used.

6. Operational Feasibility

Area	Assessment
Target users and stakeholders	ITDD staff and Database Management Division personnel as direct users; source-department data owners and the city Data Protection Officer as approving stakeholders; public open-data users as indirect beneficiaries.
Intended operating environment	An office workstation or group laptop running the application locally in a browser, used before a release decision rather than as part of any live publishing pipeline.
Availability of participants	Open and already contacted. Quezon City publishes a Request Information service and ITDD lists data extraction as a named service, so a defined channel exists, and the group has reached an identified contact through it. What is not yet known is the response time and the granularity the city will release.
User needs and benefits	A documented way to compare release options before publication, separate evidence on quality, residual risk, and utility, a repeatable record of why a particular granularity was chosen, and early identification of metadata gaps that could mislead public users.
Devices and facilities	Ordinary laptop with 8 GB RAM, a modern browser, and a restricted storage location. No procurement required.
Usability and accessibility	Risk and utility presented as a visible trade-off rather than a single pass or fail verdict, plain-language explanations of what k means, readable colour contrast, and keyboard-operable controls. Every recommendation shows the assumptions behind it.
Training and support	A short user guide, a walkthrough of the release-readiness report, and administrator documentation for adding hierarchies and datasets.
Maintenance	Generalisation hierarchies need updating as category dictionaries change, and threshold policy needs review as guidance evolves. Both are configuration rather than code.
Adoption challenges	The largest is that a system output could be mistaken for legal clearance to publish. The system therefore states in the report itself that it is decision support and does not replace the city Data Protection Officer. A second challenge is that ITDD may not have time to engage a student project; the study is designed so that the core evaluation can still be completed on a fallback dataset.

Operational finding. The operating environment is undemanding and the deliverable addresses a decision the city genuinely has to make before every release. Operational feasibility is strong because a documented request channel and a named responsible department already exist, and because no staff interviews or usability testing inside the city are required during the approved study.

7. Economic Feasibility



Expense	Estimated Cost	Purpose	Possible Alternative
Software licences	PHP 0	Python, pandas, Streamlit, SQLite, ARX or pycanon	All are open source; no paid tool is on the critical path
APIs and subscriptions	PHP 0	None required by design	Not applicable, since no external service is used
Cloud hosting and storage	PHP 0 to 600	Optional demonstration host for a public benchmark dataset only	Local demonstration; the restricted extract is never hosted
Dataset acquisition or annotation	PHP 0	Public request service carries no fee; no annotation is required	Public benchmark microdata if the request is refused
Hardware or equipment	PHP 0	Existing group laptops	School laboratory workstation if a laptop fails
Transportation and data collection	PHP 1,800	Request submission and follow-up visits to the QC Hall, estimated at three trips for two members	The online Request Information service removes most travel if it is accepted
Printing and documents	PHP 500	Formal request letter and proposal packet	Submit digitally where the city accepts it
Testing and deployment	PHP 600	Evaluation session incidentals	Remote evaluation with faculty and student testers
Maintenance	PHP 0	Handled within the study period	Not applicable during the thesis term
Contingency expenses	PHP 1,000	Follow-up visits or a change of target dataset	Reduce trips by using the online request channel
Total estimated cost	PHP 3,900 to 4,500	Working budget range	The lower bound applies if the online request channel is accepted and no hosting is used

Economic finding. One of the least expensive of the five titles, and cheaper still if the city accepts an online request, since travel is the only material line item. Cost is not a constraint on this title.

8. Schedule Feasibility

Activity	Estimated Duration	Dependency or Control
Requirements gathering	Weeks 1 to 2	Includes public dashboard surface inspection, which needs nobody's approval and starts immediately
Literature and related-study review	Weeks 1 to 5	Independent of data access; k-anonymity literature is mature and accessible
Dataset acquisition or data collection	Weeks 1 to 5	Hard decision point at Week 5. The city request is submitted in Week 2; if no approved extract exists by Week 5, switch to PSA microdata or the public benchmark
Data cleaning, annotation, and preprocessing	Weeks 5 to 7	Depends on the received schema and dictionary
System design	Weeks 2 to 5	Generalisation hierarchies drafted against the public dashboard fields while the request is pending
Algorithm or model development	Weeks 4 to 9	Suppression baseline first, then Mondrian, validated on the public benchmark before the real extract is used
System implementation	Weeks 6 to 11	Dashboard and report generator built alongside the risk engine
Model training and validation	Weeks 9 to 12	Risk and utility measured across multiple values of k; custom Mondrian cross-checked against ARX or pycanon
Software and user testing	Weeks 11 to 14	Faculty and student testers; ITDD review only if the



		department is willing
Performance evaluation	Weeks 12 to 14	Mondrian against the fixed suppression baseline on identical risk thresholds
Revision and improvement	Weeks 14 to 16	Defect correction and report refinement
Documentation	Weeks 2 to 17	Maintained continuously; every assumption recorded as it is made
Deployment and final presentation	Weeks 17 to 18	Handover package only; no integration into any city system is permitted during the study

Activities that may cause delays. City response time is the largest and it is outside the group's control, and a government request may also need a second approval from the source department. Schema and dictionary quality is the second, since undefined categories block hierarchy construction. These are managed by submitting the request in Week 2 rather than later, by building and validating the entire pipeline on a public benchmark dataset so that no development work waits on the city, and by setting a hard Week 5 decision point with two named fallbacks. Because the method is dataset-agnostic, a fallback costs the group its local finding but not its system or its evaluation.

Schedule finding. The 18-week plan holds, and it is more robust than Plan C's because a public benchmark can substitute for the real extract at any point without invalidating the work already built.

9. Legal and Ethical Feasibility

Requirement	Planned Control
Data Privacy Act compliance	Republic Act No. 10173 applies. NPC Advisory Opinion No. 2017-027 states that removing direct identifiers does not automatically make data anonymous where singling out, linkability, or inference remains possible, so the research copy is treated as restricted personal data even after source-side de-identification. NPC Advisory No. 2025-02 on privacy engineering informs the system design.
Informed consent	Not applicable. No human participants are engaged and no identifiable private information is received. Consent is obtained only if usability testing records identifiable responses.
Research ethics approval	Exempt classification is targeted. Appendix I applies for formally authorised institutional data. The adviser decides whether ITDD as data custodian triggers Appendix D. Appendix H is expected to apply, because the system produces an automated release recommendation. No data is received before ethics clearance.
Confidentiality and anonymisation	Direct identifiers and source-system identity mappings are never received. Narrative text, exact locations, and account identifiers are excluded at source. High-risk first cases involving health details, PWD identities, QCitizen ID numbers, or beneficiary financial information are excluded from dataset selection entirely.
Intellectual property and copyright	All sources cited; library and dataset licences documented; the city retains ownership of its data and the group claims no redistribution right.
Licences	Every library on the stack carries a permissive open-source licence compatible with academic use. ARX and pycanon are used as reference cross-checks under their stated licences.
Data ownership and sharing restrictions	The group never publishes the approved record-level extract. Row-level files are kept outside Git and the thesis vault and are destroyed within thirty days after the final defence unless city terms require earlier.
Cybersecurity	Restricted storage path, checksum and schema-version recording on import, prohibited-field gate, credentials in environment variables, and audit logging. No external dataset is held, so no linkage is technically possible.
AI fairness, transparency, accountability, bias	No claim is made that a passing k value constitutes legal anonymity; the system is decision support and the report says so in its own text. Results are reported across multiple values of k rather than a single chosen threshold. Reporting bias toward barangays and populations that use the service channel more is disclosed as a limitation, since a release that looks safe for



	well-represented groups may not be for rare ones.
Possible misuse or harm	Two realistic harms. First, a system output mistaken for legal clearance to publish; controlled by explicit decision-support framing in the report itself and by the statement that it does not replace the Data Protection Officer. Second, the study documenting how to evaluate re-identification risk; controlled by never attempting re-identification, holding no external dataset to link against, and reporting only aggregate risk indicators rather than any vulnerable record.

Legal and ethical finding. This title has the strongest legal grounding of the five, because the exact NPC guidance that governs it has been identified and quoted, and because privacy is the subject of the study rather than an obstacle to it. The genuine open items are the Appendix H determination, which is expected to apply, and the city's own publication conditions, which are unknown until ITDD responds.

10. Risk Assessment and Feasibility Conclusion

Identified Risk	Likelihood	Impact	Mitigation Strategy
Dataset is unavailable or insufficient	Low	High	Request submitted in Week 2; hard Week 5 decision point; two named fallbacks, being PSA anonymised microdata then a public benchmark microdata set. The system and the full evaluation survive both.
City response is slow or requires a second approval	High	Medium	Development is validated entirely on a public benchmark, so no build work waits on the city; the request explicitly names the source department as a possible co-approver.
Extract lacks exact dates or barangays	Medium	High	Without granular quasi-identifiers there is nothing to generalise, so this is asked directly in the request letter; if refused, the study moves to a dataset or benchmark that retains them.
Volume falls below the 10,000-record floor	Low	High	Helpline 122 is citywide with coverage from 2019, and the fallback benchmarks all exceed the floor comfortably.
Technology is difficult to implement	Low	Medium	Mondrian has published pseudocode; ARX and pycanon provide reference outputs to cross-check against, so an implementation error is caught rather than defended.
Algorithm produces poor results	Low	Medium	Comparison is against a stated baseline on identical thresholds. If Mondrian does not beat fixed suppression on this dataset, that is a genuine and reportable finding, not a failed study.
Required respondents are unavailable	Low	Low	No human participants are required for the core evaluation; faculty and student testers cover usability.
Privacy or ethical concerns arise	Low	High	No direct identifiers received, no linkage attempt, no re-identification attempt, high-risk datasets excluded from selection, Exempt-targeted design, and destruction within thirty days after the defence.
System output mistaken for legal clearance	Medium	Medium	Decision-support framing stated inside the generated report, explicit statement that it does not replace the Data Protection Officer, and results reported across multiple values of k rather than as a single verdict.
Project exceeds the budget	Low	Low	Total planned spending is under PHP 4,500, almost all of it travel, which the online request channel can remove.
Project cannot be completed on time	Low	High	The pipeline is built and validated on a public benchmark independently of the city, so the worst case costs the local finding rather than the deliverable.

Conclusion for this title: FEASIBLE WITH MODIFICATIONS.

Supporting findings. To conclude, the technology, the cost, the schedule, and the legal grounding of this title are all sound, and it carries the most robust fallback structure among the group's titles, which follows from a method that is

dataset-agnostic and can be validated on public benchmark microdata, so no development work waits on the city answering. The group has also already sent its request through an identified contact in the Quezon City government, and the city publishes a documented request channel with a named responsible department, which together make this a real and open line rather than a hopeful one. However, the shape of the extract is still unconfirmed, which covers whether an approved record-level extract will be released, whether it keeps the exact dates and barangays the study needs, and whether a second approval from the source department will be required. The title may proceed once ITDD confirms an approved de-identified extract of at least 10,000 rows that retains the quasi-identifiers the study needs to generalise. Yet if that is not confirmed by Week 5, the group should move to PSA anonymised microdata, and only then to a public benchmark set, which preserves the method while costing the study its local finding.

Provisional Feasibility Score for This Title

Scores use the activity's 1 to 5 scale, where 5 is highly feasible. The group's ranking across all five titles is carried in the final comparison table.

Criterion	Score	Basis
Technical and technology feasibility	4	Familiar open-source stack; Mondrian is new to the group but has published pseudocode and reference implementations
Dataset and data feasibility	3	Documented city request channel, a named department, and a contact already engaged; extract granularity still unconfirmed
Algorithm or model feasibility	4	Mature, well-documented methods with an explicit baseline and measurable outcomes
Operational feasibility	4	Undemanding environment, a real decision need, and a contact already engaged at the city
Economic feasibility	5	Under PHP 4,500, almost entirely travel
Schedule feasibility	3	Fits 18 weeks, helped considerably by benchmark-first development
Legal and ethical feasibility	3	Strongest legal grounding of the group's titles, but Appendix H is expected to apply and city conditions are unknown
Risk manageability	4	The request is sent and the method is dataset-agnostic, so fallbacks preserve the system and the evaluation
Total score	30/40	Provisional, pending the full five-title comparison
Group's ranking	2	See the final comparison table

References

- Government Data Quality Hub. (2020). The Government Data Quality Framework. gov.uk
- LeFevre, K., DeWitt, D. J., & Ramakrishnan, R. (2006). Mondrian multidimensional k-anonymity. 22nd International Conference on Data Engineering, 25.
- National Privacy Commission. (2012). Republic Act No. 10173: Data Privacy Act of 2012. privacy.gov.ph
- National Privacy Commission. (2017). Advisory Opinion No. 2017-027: Anonymized data for marketing analytics. privacy.gov.ph
- National Privacy Commission. (2025). Advisory No. 2025-02: Guidelines on privacy engineering in systems life cycle processes. privacy.gov.ph
- Quezon City Government. (n.d.). QC Open Data Dashboard. quezoncity.gov.ph/program/qc-open-data-dashboard/
- Quezon City Government. (n.d.). Quezon City Information Technology Development Department. quezoncity.gov.ph
- Quezon City Government. (2025). How to request for public information. quezoncity.gov.ph
- Samarati, P. (2001). Protecting respondents' identities in microdata release. IEEE Transactions on Knowledge and Data Engineering, 13(6), 1010-1027.



FEU Institute of Technology
COLLEGE OF ENGINEERING • COLLEGE OF COMPUTER STUDIES

Sweeney, L. (2002). k-anonymity: A model for protecting privacy. International Journal of Uncertainty, Fuzziness and Knowledge-Based Systems, 10(5), 557-570.

Reference details and the full academic literature review must be verified and completed before final submission.



Supporting Review of Related Literature and Studies

Proposal Thesis Title

A Data Governance System for Privacy Release Readiness of Quezon City Open Data Using Mondrian k-Anonymity

Review of Related Literature (RRL/s)

The RRL helps the group understand the chosen area, the problem being addressed, and the concepts or approaches relevant to the proposed solution.

Local Literatures:

1. **Republic Act No. 10173, the Data Privacy Act of 2012**

Description / Summary (Local Literature). The law protects personal information processed by government and private organisations, and requires personal information controllers to apply reasonable organisational, physical, and technical security measures. It also created the National Privacy Commission to enforce these duties.

Relevance to the proposed study. Any record-level city dataset is personal information until it is shown otherwise. The law is the reason a release has to be assessed for identifiability before publication.

Source: https://lawphil.net/statutes/repacts/ra2012/ra_10173_2012.html

2. **Executive Order No. 2, s. 2016, Operationalizing in the Executive Branch the People's Constitutional Right to Information**

Description / Summary (Local Literature). The order established the Freedom of Information programme in the executive branch and directs government offices to give the public access to information and records. It also has a section on protecting privacy, which requires officials to fully protect the individual's right to privacy and to keep personal information in their custody secure when they provide access.

Relevance to the proposed study. It sets out the tension the proposed system resolves: government must disclose, but it must not disclose personal information. The system measures where that line falls for a specific dataset.

Source: https://lawphil.net/executive/execord/eo2016/eo_2_2016.html

3. **NPC Advisory No. 2025-02, Guidelines on Privacy Engineering in Systems Life Cycle Processes**

Description / Summary (Local Literature). The National Privacy Commission issued guidelines for building privacy into every stage of a system's life cycle, from planning and requirements through design, testing, deployment, and maintenance. It applies to new, existing, and updated systems.

Relevance to the proposed study. It supports designing privacy checks into the release process itself, which is what a release-readiness step does, instead of reviewing privacy after data is already published.

Source: https://privacy.gov.ph/wp-content/uploads/2025/12/NPC_Advisory2025-02.pdf

4. **QC Open Data Dashboard**

Description / Summary (Local Literature). Quezon City's official page describes the dashboard launched by ITDD on QC e-Services, covering six city programmes with data from 2019 onward, and states that the city will publish additional datasets.

Relevance to the proposed study. It confirms the partner, the candidate datasets, and the fact that more releases are planned, each of which will need a release decision.

Source: <https://quezoncity.gov.ph/program/qc-open-data-dashboard/>



5. **Open Data Philippines portal (data.gov.ph)**

Description / Summary (Local Literature). The national open-data portal publishes government datasets from participating agencies. When the group reviewed it on September 19, 2026, and again on September 24, 2026, it listed 175 datasets from 17 publishing agencies, no local government unit publishers, and most recent retrieval dates from 2022 to 2023.

Relevance to the proposed study. It shows that local governments are largely absent from national open data. A method that makes city releases safer and easier to justify addresses one reason for that absence.

Source: <https://data.gov.ph>

Foreign Literatures:

6. **Beyond Accuracy: What Data Quality Means to Data Consumers**

Description / Summary (Foreign Literature). Wang and Strong (1996) surveyed data users to find which qualities of data matter to them, and grouped the results into intrinsic, contextual, representational, and accessibility dimensions. They showed that data quality means more than accuracy, and depends on the task the data is used for.

Relevance to the proposed study. It is the basis for measuring the quality of a release in several dimensions, and for judging quality by whether the data can answer the questions the public will ask of it.

Source: <https://doi.org/10.1080/07421222.1996.11518099>

7. **Benefits, Adoption Barriers and Myths of Open Data and Open Government**

Description / Summary (Foreign Literature). Janssen, Charalabidis, and Zuiderwijk (2012) described the benefits governments expect from open data and the barriers that stop it, including concerns about privacy and data quality. They challenged the belief that simply publishing data is enough to create value.

Relevance to the proposed study. Privacy and quality concerns are among the barriers they name. The proposed system reduces both, so it addresses a documented reason why governments hold back data.

Source: <https://doi.org/10.1080/10580530.2012.716740>

8. **k-Anonymity: A Model for Protecting Privacy**

Description / Summary (Foreign Literature). Sweeney (2002) showed that records stripped of names can still be re-identified by linking fields such as birth date, sex, and postal code with other data. She proposed k-anonymity, which requires every record to share its quasi-identifier values with at least k-1 other records.

Relevance to the proposed study. It defines the quasi-identifier and the risk measure the system uses. It is also why removing names alone does not make a city dataset anonymous.

Source: <https://doi.org/10.1142/S0218488502001648>

9. **Mondrian Multidimensional K-Anonymity**

Description / Summary (Foreign Literature). LeFevre, DeWitt, and Ramakrishnan (2006) proposed Mondrian, a greedy algorithm that recursively splits the data across several attributes at once until every group has at least k records. Because it splits on several attributes instead of one at a time, it keeps more detail than earlier single-dimension methods.

Relevance to the proposed study. It is the algorithm named in the proposed title. It is expected to reach the same k as blanket suppression while keeping more of the data useful.

Source: <https://doi.org/10.1109/ICDE.2006.101>

10. **l-Diversity: Privacy Beyond k-Anonymity**

Description / Summary (Foreign Literature). Machanavajjhala, Kifer, Gehrke, and Venkitasubramaniam (2007) showed that k-anonymity can still leak information when every record in a group shares the same sensitive value, or when an attacker has background knowledge. They proposed l-diversity, which requires each group to contain several different sensitive values.



Relevance to the proposed study. It defines the known weakness of k-anonymity, which the system reports as a stated limitation, and it gives a second check for groups where every record has the same concern category.

Source: <https://doi.org/10.1145/1217299.1217302>

Review of Related Studies (RRS/s)

The RRS focuses on existing research, systems, applications, or projects that address a problem similar to the one the group wants to solve.

Local Studies:

1. Public FOI Requests in the Philippines: Transparency at the Expense of Privacy?

Description / Summary (Local Study). Perez (2025) analysed over 235,000 requests on the Philippine eFOI portal filed between 2016 and 2024, using Presidio, an open-source personal information detector. About 44 percent of requests contained personal information and 0.66 percent contained sensitive identifiers, even though privacy advisories were in place. Exposure was higher for some agencies and during the COVID-19 pandemic. The author recommends automated de-identification together with human oversight.

Relevance to the proposed study. It is direct Philippine evidence that government transparency platforms publish personal information at scale when there is no systematic check. The proposed system is that kind of check, applied before a city dataset is released.

Source: https://doi.org/10.1007/978-981-95-4861-3_35

2. Why are Freedom of Information Requests in the Philippines Being Denied? An Analysis using a Practice-based Approach

Description / Summary (Local Study). Perez (2020) examined 346 randomly sampled requests from the eFOI portal, which had received 11,990 requests across 366 government offices by June 2019. Agencies mostly denied requests because they did not hold the information, did not consider the request valid, or processed requests differently from one another.

Relevance to the proposed study. It shows that Philippine agencies lack consistent procedures for deciding what to release. The proposed system gives a city a consistent, documented procedure for one kind of release decision.

Source: <https://phjlis.org/index.php/phjlis/article/view/41>

3. Open Data Governance in the Time of the Pandemic

Description / Summary (Local Study). Alampay (2021) assessed open government data in the Philippines by comparing access to health information during the COVID-19 pandemic with open data available before it. The study used FOI requests and open-data platforms, and analysed openness in terms of content, people, and process.

Relevance to the proposed study. It documents the state of Philippine open data and the push for more systematic data sharing across government, which is the setting the proposed system works in.

Source: <https://doi.org/10.1145/3494193.3494255>

4. Open Data in Philippines: An Issue of Access and Awareness

Description / Summary (Local Study). Pacis (2017), in a research brief for IT for Change, reviewed Philippine open data and freedom of information initiatives. The brief identified limited internet access, limited capacity of local groups, and low awareness among government staff as barriers to open data being used.

Relevance to the proposed study. It shows that the barriers to Philippine open data are practical and institutional, not only technical. A tool that helps staff make and explain release decisions addresses the capacity side of that problem.

Source: <https://projects.itforchange.net/mavc/wp-content/uploads/2017/06/Research-Brief-Philippines.pdf>



5. **National Government Agency's Compliance on Data Privacy Act of 2012: A Case Study**

Description / Summary (Local Study). Pitogo (2019) used a qualitative case study to examine how a Philippine frontline national agency complies with the Data Privacy Act. The agency was found partially compliant, and faced a lack of awareness, a wait-and-see attitude, and time and resource constraints.

Relevance to the proposed study. It shows that Philippine government offices struggle to apply privacy law in practice. A system that turns a privacy requirement into measurable checks lowers the effort needed to comply.

Source: <https://doi.org/10.1088/1742-6596/1201/1/012021>

Foreign Studies:

6. **Open Data Quality Measurement Framework: Definition and Application to Open Government Data**

Description / Summary (Foreign Study). Vetrò et al. (2016) built a framework of indicators to measure the quality of open government datasets and applied it to Italian data. Datasets from a centralised release with quality controls were compared with municipal releases without them, and the municipal data showed weaker quality on several dimensions.

Relevance to the proposed study. It shows that city-level releases without quality controls are measurably worse, and provides indicators the system can adapt for its quality profile.

Source: <https://doi.org/10.1016/j.giq.2016.02.001>

7. **Estimating the Success of Re-identifications in Incomplete Datasets Using Generative Models**

Description / Summary (Foreign Study). Rocher, Hendrickx, and de Montjoye (2019) built a model that estimates the chance that a specific person can be correctly re-identified, even in a heavily sampled dataset. They estimated that 99.98 percent of Americans would be correctly re-identified in any dataset using 15 demographic attributes.

Relevance to the proposed study. It shows that sampling and removing names do not make a dataset anonymous, which is why the system measures risk directly instead of assuming it is low.

Source: <https://doi.org/10.1038/s41467-019-10933-3>

8. **Unique in the Crowd: The Privacy Bounds of Human Mobility**

Description / Summary (Foreign Study). De Montjoye, Hidalgo, Verleysen, and Blondel (2013) studied location records from 1.5 million mobile phone users and found that four points in space and time were enough to uniquely identify 95 percent of them. Making the data coarser reduced this uniqueness only slowly.

Relevance to the proposed study. Helpline 122 records carry a place and a time. This study shows why barangay and date fields must be treated as quasi-identifiers and generalised with care.

Source: <https://doi.org/10.1038/srep01376>

9. **A Globally Optimal k-Anonymity Method for the De-identification of Health Data**

Description / Summary (Foreign Study). El Emam et al. (2009) developed a k-anonymity algorithm that searches for the generalisation with the least information loss and compared it with earlier algorithms on real datasets. Their method lost less information than the alternatives at the same level of protection.

Relevance to the proposed study. It shows that the choice of k-anonymity algorithm changes how much useful detail survives. The proposed study makes the same kind of comparison between Mondrian and fixed suppression.

Source: <https://doi.org/10.1197/jamia.M3144>

10. **Flexible Data Anonymization Using ARX: Current Status and Challenges Ahead**

Description / Summary (Foreign Study). Prasser, Eicher, Spengler, Bild, and Kuhn (2020) described ARX, an open-source anonymisation tool that supports k-anonymity and related privacy models, risk analysis, and utility measurement, and reviewed the challenges of using such tools in practice.

Relevance to the proposed study. ARX is the reference implementation the group will use to cross-check its own Mondrian output, so an implementation error is caught rather than defended.

Source: <https://doi.org/10.1002/spe.2812>

Research and System Gap

The Philippine sources show that openness and privacy are both legal duties, that government platforms have exposed personal information at scale without systematic checks, and that agencies lack consistent release procedures and the capacity to apply privacy law. The international sources show that removing names does not make data anonymous, that place and time fields are especially identifying, that k-anonymity and Mondrian can reduce risk while keeping detail, and that unmanaged municipal releases are measurably lower in quality.

The gap. No published work applies these measurements to a Philippine local government open-data release. No Philippine study compares a smarter anonymisation method against the blanket suppression that data owners actually use, on real city data, while also measuring quality. This study fills that gap with an auditable, reusable release-readiness procedure.

Refined Proposed Title and Solution

A Data Governance System for Privacy Release Readiness of Quezon City Open Data Using Mondrian k-Anonymity (16 words). The title is kept, because the review confirmed Mondrian as the right method for the comparison the study makes.

References for the RRL and RRS

- Alampay, E. A. (2021). Open data governance in the time of the pandemic. *Proceedings of the 14th International Conference on Theory and Practice of Electronic Governance*, 455–462.
<https://doi.org/10.1145/3494193.3494255>
- de Montjoye, Y.-A., Hidalgo, C. A., Verleysen, M., & Blondel, V. D. (2013). Unique in the crowd: The privacy bounds of human mobility. *Scientific Reports*, 3, 1376. <https://doi.org/10.1038/srep01376>
- El Emam, K., Dankar, F. K., Issa, R., Jonker, E., Amyot, D., Cogo, E., et al. (2009). A globally optimal k-anonymity method for the de-identification of health data. *Journal of the American Medical Informatics Association*, 16(5), 670–682. <https://doi.org/10.1197/jamia.M3144>
- Executive Order No. 2, s. 2016. (2016). Operationalizing in the executive branch the people's constitutional right to information. https://lawphil.net/executive/execord/eo2016/eo_2_2016.html
- Janssen, M., Charalabidis, Y., & Zuidewijk, A. (2012). Benefits, adoption barriers and myths of open data and open government. *Information Systems Management*, 29(4), 258–268.
<https://doi.org/10.1080/10580530.2012.716740>
- LeFevre, K., DeWitt, D. J., & Ramakrishnan, R. (2006). Mondrian multidimensional k-anonymity. *Proceedings of the 22nd International Conference on Data Engineering*, 25. <https://doi.org/10.1109/ICDE.2006.101>
- Machanavajjhala, A., Kifer, D., Gehrke, J., & Venkatasubramanian, M. (2007). l-Diversity: Privacy beyond k-anonymity. *ACM Transactions on Knowledge Discovery from Data*, 1(1), 3.
<https://doi.org/10.1145/1217299.1217302>
- National Privacy Commission. (2025). NPC Advisory No. 2025-02: Guidelines on privacy engineering in systems life cycle processes. https://privacy.gov.ph/wp-content/uploads/2025/12/NPC_Advisory2025-02.pdf
- Open Data Philippines. (n.d.). data.gov.ph. Retrieved September 19, 2026, from <https://data.gov.ph>
- Pacis, J. (2017). Open data in Philippines: An issue of access and awareness [Research brief]. IT for Change. <https://projects.itforchange.net/mavc/wp-content/uploads/2017/06/Research-Brief-Philippines.pdf>



- Perez, P. J. (2020). Why are freedom of information requests in the Philippines being denied? An analysis using a practice-based approach. *Philippine Journal of Librarianship and Information Studies*.
<https://phjlis.org/index.php/phjlis/article/view/41>
- Perez, P. J. (2025). Public FOI requests in the Philippines: Transparency at the expense of privacy? In *Intelligence and equity: Shaping the future of knowledge (Lecture Notes in Computer Science, pp. 417–426)*. Springer.
https://doi.org/10.1007/978-981-95-4861-3_35
- Pitogo, V. (2019). National government agency's compliance on Data Privacy Act of 2012: A case study. *Journal of Physics: Conference Series*, 1201, 012021. <https://doi.org/10.1088/1742-6596/1201/1/012021>
- Prasser, F., Eicher, J., Spengler, H., Bild, R., & Kuhn, K. A. (2020). Flexible data anonymization using ARX: Current status and challenges ahead. *Software: Practice and Experience*, 50(7), 1277–1304.
<https://doi.org/10.1002/spe.2812>
- Quezon City Government. (n.d.). QC Open Data Dashboard. <https://quezoncity.gov.ph/program/qc-open-data-dashboard/>
- Republic Act No. 10173. (2012). Data Privacy Act of 2012.
https://lawphil.net/statutes/repacts/ra2012/ra_10173_2012.html
- Rocher, L., Hendrickx, J. M., & de Montjoye, Y.-A. (2019). Estimating the success of re-identifications in incomplete datasets using generative models. *Nature Communications*, 10, 3069. <https://doi.org/10.1038/s41467-019-10933-3>
- Sweeney, L. (2002). k-Anonymity: A model for protecting privacy. *International Journal of Uncertainty, Fuzziness and Knowledge-Based Systems*, 10(5), 557–570. <https://doi.org/10.1142/S0218488502001648>
- Vetrò, A., Canova, L., Torchiano, M., Minotas, C. O., Iemma, R., & Morando, F. (2016). Open data quality measurement framework: Definition and application to open government data. *Government Information Quarterly*, 33(2), 325–337. <https://doi.org/10.1016/j.giq.2016.02.001>
- Wang, R. Y., & Strong, D. M. (1996). Beyond accuracy: What data quality means to data consumers. *Journal of Management Information Systems*, 12(4), 5–33. <https://doi.org/10.1080/07421222.1996.11518099>



TARGET TITLE 4 OF 5

Development of a Fuzzy Matching and Data Quality Scoring Model for Detecting Potential Duplicate and Incomplete Child Immunization Records at a Barangay Health Center in Mandaluyong City

1. Target Thesis Title

Development of a Fuzzy Matching and Data Quality Scoring Model for Detecting Potential Duplicate and Incomplete Child Immunization Records at a Barangay Health Center in Mandaluyong City

Thesis specialization: Data Science

Primary technique: Data Governance and Ethics, from the approved Data Science technique list

Research domain: Data science, health informatics, data quality, record linkage, and immunization information management

Proponents: Dawn Andrei Pamesa (Leader), Rogelio Porras, John Ashley Alba, Ralph Blanco

2. Project Overview

Background of the problem. Barangay health centers maintain child immunization records used to monitor vaccine completion and prepare program reports. Missing vaccine dates, invalid ages, inconsistent entries, and duplicate records can make it harder for health workers to identify children needing follow up and can distort coverage counts.

Specific problem. Existing health information policies and research recognize data quality and record matching problems, but a barangay health center may not have a simple, explainable tool that flags incomplete child immunization records and possible duplicate entries before staff prepare reports or conduct follow up.

Target users and beneficiaries. Barangay health workers, public health nurses, health center record staff, city health office immunization personnel, and academic evaluators.

Proposed solution. A Python based data science prototype that applies rule based validation, fuzzy duplicate matching, and a weighted Data Quality Score to approved de identified or simulated child immunization records. It produces review flags and corrective recommendations; staff remain responsible for confirming any record.

Main objectives. Define a child immunization data dictionary; assess completeness, validity, consistency, and uniqueness of approved data; flag possible duplicate records for human review; calculate a Data Quality Score; and create a simple report or dashboard for health workers.

Expected output. A de identified or simulated child immunization dataset, documented validation rules, possible duplicate flags, Data Quality Score reports, a prototype dashboard or report, and recommendations for record review.

Intended contribution. A low cost, locally relevant data science prototype for improving the review of child immunization records at one barangay health center. It supports data review and reporting; it does not replace an electronic medical record system, verify vaccination status, or make clinical decisions.

3. Technical and Technology Feasibility

Area	Proposed Choice	Feasibility and Alternative
------	-----------------	-----------------------------



Programming languages	Python	Covers validation, matching, scoring, and the report in one language.
Frameworks and libraries	pandas, RapidFuzz or jellyfish for Jaro-Winkler and Levenshtein similarity, Streamlit	All open source and widely documented.
Database and storage	SQLite or encrypted CSV files on approved storage	No server needed for a single health center.
APIs and third-party services	None	Nothing on the critical path depends on an outside service.
Platform	Streamlit report or dashboard, run locally	Can be demonstrated offline during the defense.
Hardware and equipment	A laptop or school workstation	No specialised hardware is needed.
Architecture and integration	Import of a de-identified or simulated file, then rule validation, fuzzy duplicate matching, Data Quality Scoring, and a report	Reads a dataset export only and never alters the official immunization register.
Security and maintenance	Restricted access, secure local storage, no direct identifiers	Health center staff retain every correction decision.
Technical expertise	Python and pandas are in hand; string similarity libraries are simple to use.	Threshold tuning is the least familiar step and is documented in the record linkage literature.
Limitations and alternatives	Matching quality depends on which fields are approved.	If real data is not approved, the simulated dataset is used.

Technical finding. The stack is simple, free, and familiar, and the tool runs on a single laptop.

4. Dataset and Data Feasibility

Requirement	Current Assessment
Dataset name and description	Child immunization records: coded child ID, date of birth, sex, vaccine dose dates, and approved matching fields such as simulated name tokens and a guardian token.
Source, owner, and provider	A barangay health center in Mandaluyong City, which owns the records. The specific health center is still being identified, so no request has been sent yet.
Public link	None. Records are held by the health center.
Data type and file format	Tabular export, CSV or XLSX.
Estimated size	To be confirmed once a health center agrees to take part. The study needs at least 10,000 records, which is difficult to reach at a single health center, and whether simulated records count toward that floor is still to be confirmed with the professor.
Features and attributes	Coded child ID, birth date, sex, vaccine and dose dates, and matching tokens.
Labels and target variables	Deliberately injected errors and labelled simulated duplicate pairs, used to measure detection.
Accessibility and collection	Written site permission for any real data; otherwise a realistic simulated dataset built from the data dictionary.
Permission and licensing	Official records stay under the health center's control; the team may not copy, redistribute, or publish patient data.
Quality, completeness, and bias	Missing vaccine dates, invalid ages, inconsistent entries, and duplicates are the very problems measured.
Preprocessing and storage	De-identification checklist; no free text, screenshots, or direct child or guardian identifiers.



Training, validation, testing	Rules and thresholds are tuned on one part of the simulated data and tested on another.
Alternative data plan	The simulated child immunization dataset is the documented fallback.

Dataset finding. Feasible with the simulated-data plan. Real records would strengthen the study but require written permission and a stricter ethics route.

5. Algorithm or Model Feasibility

Evaluation	Approach 1	Approach 2	Approach 3
Name	Rule based validation	Fuzzy duplicate matching	Weighted data quality scoring
Purpose	Detect missing, invalid, and inconsistent immunization fields	Flag records that may refer to the same child	Summarize the quality of each record and dataset
Method	Required field, date sequence, age range, and format rules	Jaro Winkler or Levenshtein similarity using approved de identified or simulated child attributes	Configured weights and documented thresholds based on validation results
Data needs	Fields such as coded child ID, date of birth, sex, vaccine dose dates, and validation rules	At least two comparable approved fields, such as simulated name tokens, birth date, sex, and guardian token	Validation and duplicate matching results
Output	Error type and corrective recommendation	Possible duplicate flag with similarity reason for human review	Record and dataset Data Quality Scores
Evaluation	Percentage of correctly detected test errors	Precision, recall, and F1 score on labelled simulated duplicate pairs	Clarity, consistency, and usefulness of score categories
Limitations	Rules must be reviewed by staff and follow the approved data dictionary	A flag is not proof of a duplicate and must never trigger an automatic merge	Scores guide review and do not establish vaccination status or clinical eligibility
Recommendation	Required core method	Required with manual review	Required reporting method

6. Operational Feasibility

Area	Assessment
Users and stakeholders	Barangay health workers, public health nurses, record staff, city health office immunization personnel, faculty, student developers, and evaluators.
Environment and facilities	A laptop or school workstation, a browser for the prototype, and secure local storage for a non identifiable child immunization test dataset.
Benefits	Earlier identification of incomplete or inconsistent child immunization entries, possible duplicate flags, and clearer prioritization of records for staff review.
Training	Short orientation on interpreting validation flags, similarity scores, and Data Quality Scores; health staff retain judgment over every correction.
Workflow fit	The prototype reviews a dataset export or simulated file and does not automatically alter the official immunization register or patient record system.
Support	The student team maintains the prototype during the study; future use requires health center approval and an assigned responsible staff member.



7. Economic Feasibility

Expense	Estimated cost	Purpose	Alternative
Software licenses	PHP 0	Python and open source libraries	Use free open source tools
Data storage	PHP 0 to 500	Encrypted local storage or approved drive	Use password protected school storage if permitted
Hosting	PHP 0 to 500	Optional Streamlit demonstration	Run locally during defense
Printing and documentation	PHP 500 to 1,500	Letters, consent documents, and thesis printing	Use digital documents where allowed
Transportation	PHP 500 to 2,000	Visits to the approved health center	Use online meetings where accepted
Contingency	PHP 500 to 1,000	Minor testing and presentation costs	Keep a small reserve
Estimated total	PHP 1,500 to 5,500	Undergraduate prototype and documentation	Remain feasible within a low budget

8. Schedule Feasibility

Period	Activities
Weeks 1 to 2	Confirm title, research site, data permission, ethics requirements, and literature review.
Weeks 2 to 4	Create a simulated or de identified child immunization dataset, data dictionary, validation rules, and duplicate-pair labels.
Weeks 4 to 6	Develop validation and fuzzy matching prototype; test data-error detection and possible duplicate flags.
Weeks 6 to 8	Build the Data Quality Score report or dashboard; conduct usability feedback with authorized health workers if approved.
Weeks 8 to 10	Analyze results, revise the prototype, document limitations, and prepare thesis chapters.
Weeks 10 to 12	Finalize testing, defense materials, source citations, and recommendations.

Possible delays. Permission processing, staff availability, privacy review, and refinement of child immunization validation rules may delay the study. Manage them by using a simulated-data fallback, preparing an early prototype, keeping the questionnaire short, and avoiding features that require access to identifiable records.

9. Legal and Ethical Feasibility

Topic	Assessment
Privacy and consent	Use only simulated or de identified child immunization data. Obtain written site permission before observing workflows or asking staff to answer surveys.
Data Privacy Act	Treat Republic Act No. 10173 and its IRR as RRL legal sources. The prototype must use purpose limitation, restricted access, and appropriate safeguards.
Ethics review	Confirm school requirements before any staff survey, interview, observation, or data access. The project must remain minimal risk.
Ownership and access	Official health center records remain under the health center control. The student team may not copy, redistribute, or publish patient data.



Automated decisions	The tool only flags possible data problems. It does not determine vaccination status, treatment, eligibility, or any other decision affecting a child.
Documentation	Keep a child immunization data dictionary, rule list, consent or permission records, and summary results without identifiers.

10. Risk Assessment and Feasibility Conclusion

Risk	Likelihood	Impact	Mitigation
Permission for real data is not granted	Medium	Medium	Use a realistic simulated child immunization dataset and focus on the prototype and workflow assessment.
Privacy risk from accidental identifiers	Low	High	Use a strict de identification checklist; do not collect free text, screenshots, or direct child or guardian identifiers.
Rules do not match the immunization workflow	Medium	Medium	Review validation rules with authorized health workers and document approved changes.
Too many duplicate flags	Medium	Medium	Use conservative thresholds and require human review before identifying a true duplicate.
Limited staff availability	Medium	Medium	Use a short questionnaire and schedule one focused review session.
Prototype scope becomes too large	Medium	Medium	Keep the first version to validation, fuzzy matching, Data Quality Scoring, and a simple report.

Conclusion for this title: FEASIBLE WITH MODIFICATIONS.

Supporting findings. The Python tools, skills, hardware, and budget are manageable. The study may proceed after the team confirms written approval for any real workflow assessment or adopts the documented simulated-data fallback. The prototype must remain an explainable, nonclinical data quality and possible-duplicate assessment tool.

Provisional Feasibility Score for This Title

Scores use the activity's 1 to 5 scale, where 5 is highly feasible. The group's ranking across all five titles is carried in the final comparison table.

Criterion	Score	Basis
Technical and technology feasibility	5	Simple, familiar, free stack
Dataset and data feasibility	3	Simulated fallback is in the group's control; real records need permission
Algorithm or model feasibility	4	Established rule and fuzzy matching methods
Operational feasibility	3	Needs health worker time for rules and feedback
Economic feasibility	5	PHP 1,500 to 5,500
Schedule feasibility	4	12-week plan with a simulated-data fallback
Legal and ethical feasibility	4	Child health data is sensitive, but only simulated or de-identified records are used
Risk manageability	4	Every listed risk has a stated mitigation
Total score	32/40	
Group's ranking	5	See the final comparison table



Supporting Review of Related Literature and Studies

Proposal Thesis Title

Development of a Fuzzy Matching and Data Quality Scoring Model for Detecting Potential Duplicate and Incomplete Child Immunization Records at a Barangay Health Center in Mandaluyong City

Review of Related Literature (RRL/s)

The RRL helps the group understand the chosen area, the problem being addressed, and the concepts or approaches relevant to the proposed solution.

Local Literatures:

1. **Republic Act No. 10173 Data Privacy Act of 2012**
Description / Summary (Local Literature). Section 11 sets privacy principles and Section 20 requires safeguards. It requires restricted, lawful handling of child immunization data and supports the use of simulated or de identified records.
2. **Data Privacy Act Implementing Rules and Regulations**
Description / Summary (Local Literature). Section 25 requires organizational, physical, and technical safeguards. It supports access restriction, secure storage, and a clear data handling plan for the prototype.
3. **Republic Act No. 11223 Universal Health Care Act**
Description / Summary (Local Literature). The law supports integrated health services. Reliable immunization information is important for service continuity, monitoring, and local health reporting.
4. **Philippine eHealth Strategic Framework and Plan**
Description / Summary (Local Literature). The framework provides Philippine direction for safe and useful health information management, supporting practical digital improvements in primary care.

Foreign Literatures:

5. **WHO Immunization Agenda 2030**
Description / Summary (Foreign Literature). The strategy emphasizes equitable immunization and the use of good information to identify missed children and improve program performance.
6. **WHO Monitoring and Evaluating Digital Health Interventions**
Description / Summary (Foreign Literature). The guide supports clear indicators and evidence for evaluating the prototype through error detection, duplicate matching results, and staff feedback.
7. **WHO Data Quality Review Toolkit**
Description / Summary (Foreign Literature). The toolkit describes routine checks for completeness, consistency, and reporting quality. Its ideas can be adapted into child immunization validation rules.
8. **Methods and Dimensions of Electronic Health Record Data Quality Assessment**
Description / Summary (Foreign Literature). This work identifies practical data quality dimensions such as completeness, correctness, concordance, plausibility, and currency that can guide the scoring model.
9. **A Harmonized Data Quality Assessment Terminology and Framework for the Secondary Use of Electronic Health Record Data**
Description / Summary (Foreign Literature). The framework organizes data quality checks into conformance, completeness, and plausibility, helping convert record review into measurable rules.



Additional Literature Also Consulted

Source	Type	Summary and relevance
A Framework for Health Information Governance A Scoping Review	Foreign Literature	The review shows that data quality needs both technical controls and human accountability, supporting human review of every duplicate flag.

Review of Related Studies (RRS/s)

The RRS focuses on existing research, systems, applications, or projects that address a problem similar to the one the group wants to solve.

Local Studies:

- Understanding Adoption of Electronic Medical Records During a Health Emergency**
Description / Summary (Local Study). Philippine usage logs from rural facilities showed that EMR feature use varies by user and context; installing a system alone does not ensure reliable or consistently entered records.
- Strengthening Health Information Systems Policies Gaps and a Framework for Reform**
Description / Summary (Local Study). Philippine evidence identifies fragmented data management, limited standardization, and interoperability issues that require practical local improvements.
- Feedback Workshop and Random Monitoring as Quality Assurance Interventions**
Description / Summary (Local Study). A Philippine study found that feedback, training, and monitoring can improve electronic record entries, supporting score based review of immunization records.
- Implementing Electronic Health Records in Philippine Primary Care Settings Mixed Methods Pilot Study**
Description / Summary (Local Study). The pilot found that workflow fit, training, connectivity, technical performance, and privacy design influence whether primary care staff accept EHR systems.

Foreign Studies:

- Optimized Dual Threshold Entity Resolution for Electronic Health Record Databases**
Description / Summary (Foreign Study). The study evaluates record linkage methods for finding records that may belong to the same patient. It supports fuzzy matching as a method for generating possible duplicate flags rather than automatic merges.
- Electronic Health Record Data Quality Assessment and Tools A Systematic Review**
Description / Summary (Foreign Study). The review found common quality dimensions but no single standard assessment method, creating an opportunity for a transparent local scoring tool.
- Digital Health Data Quality Issues A Systematic Review**
Description / Summary (Foreign Study). The review identifies recurring missing, inaccurate, inconsistent, and duplicated data problems that can weaken reporting and later analysis.
- Linking Patient Records at Scale with a Hybrid Approach Combining Contrastive Learning and Deterministic Rules**
Description / Summary (Foreign Study). This recent patient linkage study shows that combining rules with matching models can improve record linkage. The proposed undergraduate system uses the simpler, more explainable rule and fuzzy matching approach.

Research and System Gap

The legal, immunization, and technical literature supports secure data handling, defined quality checks, and human accountability for local immunization information.

The gap. The reviewed frameworks explain good data quality and immunization monitoring, but do not provide a small Philippine barangay health center with an explainable tool for jointly flagging incomplete and potentially duplicate child immunization records.

The RRL supports four measurable dimensions: completeness, validity, consistency, and uniqueness. These become rule checks, fuzzy matching features, and the Data Quality Score.

References for the RRL and RRS

- Acacio Claro, P. J., et al. (2024). Understanding adoption of electronic medical records during a health emergency. *Acta Medica Philippina*, 58(22). <https://doi.org/10.47895/amp.vi0.8911>
- Conjares, M. N. T. M., et al. (2025). Strengthening health information systems policies gaps and a framework for reform. *PIDS Discussion Paper 2025-31*. <https://doi.org/10.62986/dp2025.31>
- Elepano, A., et al. (2025). Implementing electronic health records in Philippine primary care settings: Mixed methods pilot study. *JMIR Medical Informatics*, 13, e63036. <https://doi.org/10.2196/63036>
- Jin, D., et al. (2023). Digital health data quality issues: Systematic review. *Journal of Medical Internet Research*, 25, e42615. <https://www.jmir.org/2023/1/e42615>
- Kahn, M. G., et al. (2016). A harmonized data quality assessment terminology and framework for the secondary use of electronic health record data. *Journal of the American Medical Informatics Association*, 23(6), 1049-1059. <https://doi.org/10.1093/jamia/ocv212>
- Lewis, A. E., et al. (2023). Electronic health record data quality assessment and tools: A systematic review. *Journal of the American Medical Informatics Association*, 30(10), 1730-1740. <https://doi.org/10.1093/jamia/ocad120>
- National Privacy Commission. (2021). Implementing rules and regulations of the Data Privacy Act of 2012 as amended. https://privacy.gov.ph/wp-content/uploads/2023/06/IRR_RA-10173-as-amended.pdf
- Republic Act No. 10173. (2012). Data Privacy Act of 2012. https://lawphil.net/statutes/repacts/ra2012/ra_10173_2012.html
- Republic Act No. 11223. (2019). Universal Health Care Act. https://lawphil.net/statutes/repacts/ra2019/ra_11223_2019.html
- Republic Act No. 12254. (2025). E-Governance Act. https://lawphil.net/statutes/repacts/ra2025/ra_12254_2025.html
- Sweeney, C., et al. (2013). Optimized dual threshold entity resolution for electronic health record databases: Training set size and active learning. *Journal of the American Medical Informatics Association*, 21(1), 113-121. <https://pmc.ncbi.nlm.nih.gov/articles/PMC3900213/>
- Weiskopf, N. G., & Weng, C. (2013). Methods and dimensions of electronic health record data quality assessment. *Journal of the American Medical Informatics Association*, 20(1), 144-151. <https://doi.org/10.1136/amiajnl-2011-000681>
- World Health Organization. (2017). Data quality review toolkit. <https://www.who.int/publications/i/item/9789241512725>
- World Health Organization. (2020). Immunization agenda 2030: A global strategy to leave no one behind. <https://www.who.int/publications/m/item/immunization-agenda-2030-a-global-strategy-to-leave-no-one-behind>
- World Health Organization. (2021). Global strategy on digital health 2020-2025. <https://www.who.int/publications/i/item/9789240020924>



FEU Institute of Technology
COLLEGE OF ENGINEERING • COLLEGE OF COMPUTER STUDIES



TARGET TITLE 5 OF 5

Demand Forecasting and Dynamic Price Optimization for Milk Retail Products Considering Remaining Shelf Life Using XGBoost

1. Target Thesis Title

Demand Forecasting and Dynamic Price Optimization for Milk Retail Products Considering Remaining Shelf Life Using XGBoost

Thesis specialization: Data Science

Primary technique: Predictive Analytics, from the approved Data Science technique list, supported by Optimization

Research domain: Machine Learning, Demand Forecasting, Dynamic Pricing, Retail Analytics, and Optimization

Proponents: Dawn Andrei Pamesa (Leader), Rogelio Porras, John Ashley Alba, Ralph Blanco

2. Project Overview

Background of the problem. Retail businesses that sell perishable products such as fresh milk, bread, yogurt, meat, and other short shelf-life goods face a unique pricing problem. Demand changes over time, while the opportunity to sell these products decreases as they approach expiration. Historical sales, pricing, inventory, and shelf-life data can be analyzed using machine learning to estimate future demand and support pricing decisions before products become unsellable.

Specific problem. General retail demand forecasting and price optimization methods may focus mainly on sales history, price, promotions, and seasonality. For milk products, retailers must also consider remaining inventory and the limited time available to sell each item. Poor pricing decisions may result in unsold products, unnecessary markdowns, or missed revenue opportunities. The study will investigate how demand forecasts and remaining shelf-life information can be combined to support dynamic pricing decisions.

Target users and beneficiaries. The target users are retail store owners, managers, pricing staff, and inventory personnel who handle milk products and make decisions about product pricing, stock movement, and markdowns.

Proposed solution. The study will develop a data-driven decision-support system for milk retail products; Machine learning models will analyze historical sales and product information to forecast demand. The study will compare models such as Linear Regression, Random Forest, and XGBoost using appropriate forecasting metrics. The selected model will estimate demand at candidate prices. A price optimization component will then evaluate those prices while considering predicted demand, product cost, remaining inventory, and remaining shelf life. The results will be presented through an interactive dashboard.

Main objectives. The project aims to analyze historical sales and pricing patterns of milk retail products, develop and compare machine learning models for demand forecasting, and examine how price and remaining shelf life relate to product demand, product cost, inventory, and shelf-life constraints. It also aims to create a dashboard that presents demand forecasts, product conditions, and pricing recommendations in an understandable format.

Expected output. The expected output is a retail analytics and decision-support system for milk products. The system will provide historical sales and price visualizations, demand forecasts, inventory and shelf-life information, candidate price evaluations, and recommended pricing decisions. The study will also report the performance of the selected machine learning models using metrics such as MAE, RMSE, and MAPE.

Intended contribution. The study will provide a more specific application of machine learning and optimization to retail pricing by focusing on products with limited shelf lives. Instead of treating pricing as a general retail problem,



the proposed approach will investigate how remaining shelf life and inventory conditions can be incorporated into demand-based pricing decisions. The project may help retailers make more informed decisions about milk inventory while demonstrating a practical Data Science application of demand forecasting and optimization.

3. Technical and Technology Feasibility

Date Submitted:

4. Dataset and Data Feasibility

Requirement	Current assessment
Dataset	Historical transaction data for perishable retail products containing product, date, selling price, quantity sold, inventory, and shelf-life or expiration information.
Provider	Primary data may come from a partner grocery, convenience store, bakery, or other retailer selling perishable products. Suitable public datasets may support model development when necessary.
Links	To be finalized after confirming the primary dataset and participating retailer.
Format and volume	CSV, Excel, or database records are acceptable. At least one year of historical data is preferred to capture seasonal patterns, subject to actual data availability.
Features	Date, product ID/category, selling price, quantity sold, inventory level, product cost, promotions/discounts, expiration date or remaining shelf life, and derived calendar features.
Target	The main prediction target is product demand, represented by quantity sold during a defined period. Predicted demand will serve as an input to the dynamic price optimization process.
Collection	Historical sales, inventory, pricing, and shelf-life records will be requested from the participating retailer or extracted from its available records.
Rights and privacy	Permission will be obtained before using private retail data. Personally identifiable customer information is unnecessary and will be excluded. Sensitive business information will be protected when required.
Quality and bias	Data will be examined for missing transactions, incorrect prices, duplicates, stockouts, inconsistent product records, missing expiration information, and unusual sales events.
Preprocessing	Data cleaning, missing-value handling, duplicate removal, outlier assessment, categorical encoding, time-based feature engineering, and calculation of remaining shelf life will be performed.
Validation split	A chronological train-validation-test split will prevent future information from leaking into earlier observations. Time-series cross-validation may also be evaluated.
Fallback	If complete store data are unavailable, the scope may be limited to selected perishable products with sufficient sales and shelf-life records. A suitable public dataset may support model development where appropriate.

5. Algorithm or Model Feasibility

Evaluation	Approach 1	Approach 2	Approach 3
Name	Linear Regression	Random Forest	XGBoost
Purpose	Baseline model for predicting product demand and examining price-demand	Nonlinear demand prediction using multiple retail factors	Advanced boosting model for demand forecasting



	relationships		
Data needs	Historical price, quantity sold, and basic product/time features	Historical sales with price, product, promotions, inventory, and time features	Larger structured dataset with sales, pricing, promotions, inventory, and engineered time features
Expected result	Simple and interpretable baseline	Strong performance with nonlinear retail patterns	Potentially higher predictive performance after tuning

Recommended approach. Linear Regression will serve as the baseline and will be compared with Random Forest and XGBoost for forecasting demand for selected perishable products. Input features may include historical sales, prices, inventory, promotions, calendar variables, and remaining shelf life. Model performance will be evaluated using MAE, RMSE, and MAPE. The selected model will estimate demand at candidate prices. The dynamic price optimization component will then evaluate these prices based on predicted demand, product cost, remaining inventory, and remaining shelf life under defined constraints.

6. Operational Feasibility

Area	Assessment
Users and stakeholders	Retail store owners, managers, inventory staff, and personnel responsible for pricing and managing perishable products.
Environment and facilities	A modern web browser, internet connection when deployed online, and a standard laptop or desktop computer. No specialized hardware is required.
Benefits	Provides demand forecasts and pricing analysis while considering inventory and remaining shelf life. It can support decisions involving pricing and perishable stock management.
Usability	Dashboard showing product demand, current inventory, remaining shelf life, historical sales, candidate prices, and pricing recommendations.
Training and support	User guide explaining forecasts, shelf-life information, pricing recommendations, and data updates.
Maintenance	New transactions, inventory records, and shelf-life data will be validated and added. Models may be retrained as additional historical data becomes available.
Adoption challenges	Missing expiration records, inconsistent inventory tracking, limited historical price variation, and retailer hesitation toward model-generated pricing recommendations.

7. Economic Feasibility

Expense	Estimated cost	Purpose	Alternative
Software licenses	PHP 0	Python, MySQL, and open-source machine learning libraries	Free academic/open-source tools
APIs/subscriptions	PHP 0-500	Optional external services or deployment tools	Use free-tier services or locally stored retail data
Hosting/storage	PHP 0-1,000	Hosting the web application, database, and model	Free-tier hosting or local deployment
Dataset/collection	PHP 0-500	Collection, preparation, and storage of historical sales, inventory, pricing, and shelf-life data.	Partner-store records or suitable public datasets
Hardware	PHP 0	Existing laptops or school computers for development and model training	School laboratory computers or Google Colab



Consultation/testing	PHP 0-1,000	Meetings with retail stakeholders, user testing, and system evaluation	Online meetings and local system testing
Contingency	PHP 1,000	Unexpected hosting, data collection, transportation, or service expenses	Use free services and local deployment when possible
Total	PHP 1,000-4,000	Estimated project development expenses	A mostly free and open-source setup can reduce costs

8. Schedule Feasibility

Period	Activities
--------	------------

Possible delays. Limited historical sales data, missing shelf-life or expiration records, inconsistent inventory information, and model tuning may affect the project timeline. These risks will be addressed through early dataset assessment, weekly milestones, regular backups, early baseline development, and limiting the study to products with sufficient records when necessary.

9. Legal and Ethical Feasibility

Topic	Assessment
Privacy and consent	The study will use historical retail sales and product data. Personal customer information will not be required. If private store data are used, permission will be obtained from the store, and sensitive business information will be anonymized when necessary.
Ethics review	The study is expected to involve minimal risk. Any surveys, interviews, or usability testing involving participants will follow the institution's research ethics requirements and informed consent procedures.
Ownership and licensing	Store-provided data will be used only with permission. Public datasets, software libraries, and other resources will follow their respective licenses and terms of use. All external sources will be properly cited.
Cybersecurity	User authentication, password hashing, access controls, input validation, and regular backups will be implemented. Sensitive store data and account credentials will not be publicly exposed.
Fairness and transparency	The study will document the features, preprocessing methods, model evaluation metrics, and limitations. Demand forecasting results will be evaluated using MAE, RMSE, and MAPE. Factors such as limited data, seasonal patterns, and unusual sales events will also be considered when interpreting results.
Misuse	Dynamic pricing recommendations will serve only as decision-support information and will not automatically change product prices. The system will not recommend selling products beyond their valid shelf life or expiration date. Users will be informed that demand forecasts and recommended prices are estimates based on historical data and defined optimization criteria.
Fairness and transparency	The study will document the features, preprocessing methods, model evaluation metrics, optimization criteria, and limitations. Pricing recommendations will clearly show that they are model-based estimates influenced by historical demand, inventory, price, and remaining shelf life.

10. Risk Assessment and Feasibility Conclusion

Risk	Likelihood	Impact	Mitigation
------	------------	--------	------------



Insufficient sales, inventory, or shelf-life data	Medium	High	Assess data availability before model development. Prioritize products with sufficient sales, price, inventory, and shelf-life records. Reduce the product scope or use an appropriate public dataset for model development if necessary.
Technology difficult to implement	Low	Medium	Use familiar technologies such as Python, MySQL, Scikit-learn, XGBoost, and a simple web framework. Develop each component separately before system integration.
Algorithms produce poor results	Medium	High	Compare Linear Regression, Random Forest, and XGBoost. Perform feature engineering and hyperparameter tuning, and evaluate models using MAE, RMSE, and MAPE.
Respondents unavailable	Low	Low	Schedule usability testing early. If retail staff are unavailable, qualified evaluators may be considered according to the study's approved evaluation plan.
Insufficient price variation	Medium	High	Evaluate historical price variation during dataset assessment. Limit optimization experiments to products with enough observed price

Conclusion for this title: FEASIBLE WITH MODIFICATIONS.

Supporting findings. The proposed Machine Learning-Based Demand Forecasting and Dynamic Price Optimization for Perishable Retail Products project is technically, operationally, economically, and academically feasible using available data science tools and standard computing resources. The main feasibility concern is obtaining sufficient historical retail data containing sales, prices, inventory levels, and shelf-life or expiration information. Product costs, promotions, and historical price variation can further strengthen the demand forecasting and optimization components. The team should assess data availability early and limit the study to perishable products with sufficient records when necessary. With proper data preparation, model evaluation, optimization, system development, and project scheduling, the study can be completed within the proposed timeframe.

Provisional Feasibility Score for This Title

Scores use the activity's 1 to 5 scale, where 5 is highly feasible. The group's ranking across all five titles is carried in the final comparison table.

Criterion	Score	Basis
Technical and technology feasibility	5	Ordinary open-source web and analytics stack
Dataset and data feasibility	3	No partner store confirmed; public retail datasets are a fallback
Algorithm or model feasibility	4	Three established approaches compared on the same metrics
Operational feasibility	3	Usable deliverable, but no participating store or named evaluator yet
Economic feasibility	5	PHP 1,000 to 4,000, the lowest of the five titles
Schedule feasibility	4	16-week plan with no external authorisation on the critical path
Legal and ethical feasibility	4	No personal customer data; store permission still needed
Risk manageability	3	Data and price-variation risks are real but have stated fallbacks
Total score	31/40	
Group's ranking	3	See the final comparison table



Proposal Thesis Title

Demand Forecasting and Dynamic Price Optimization for Milk Retail Products Considering Remaining Shelf Life Using XGBoost

Review of Related Literature (RRL/s)

The RRL helps the group understand the chosen area, the problem being addressed, and the concepts or approaches relevant to the proposed solution.

Local Literatures:

1. **Philippines: Dairy and Products Annual**

Description / Summary (Local Literature). The U.S. Department of Agriculture's Foreign Agricultural Service office in Manila reports that the Philippines produces only about one percent of its dairy requirement and imports the rest. Local production supplies only about 20 percent of ready-to-drink milk, with imported UHT milk covering most of the demand, and demand for ready-to-drink milk is forecast to grow 3 percent in 2025, driven by the Milk Feeding Program and the expanding food service sector.

Relevance to the proposed study. It shows that milk is a steady and growing retail category in the Philippines, and that most ready-to-drink milk sold in the country is imported stock with a limited shelf life, so milk left unsold before expiry is a direct loss to the retailer. This supports milk as the product focus of the study.

Source:

https://apps.fas.usda.gov/newgainapi/api/Report/DownloadReportByFileName?fileName=Dairy+and+Products+Annual_Manila_Philippines_RP2024-0038.pdf

2. **Food Loss and Waste in the Philippines: A Literature Review**

Description / Summary (Local Literature). Barrion, Calayag, Nguyen-Orca, and Melo-Rijik (2023) reviewed the current state of food loss and waste in the Philippines along the supply chain and the Food Waste Reduction Act. Among the recommendations for the food industry, they list strict first-in, first-out practice to minimise spoilage.

Relevance to the proposed study. It is Philippine evidence that food waste is a recognised problem and that managing stock by its age is an accepted way to reduce it. The proposed study goes further by using remaining shelf life to set a price before the product becomes waste.

Source: [https://doi.org/10.26656/fr.2017.7\(6\).127](https://doi.org/10.26656/fr.2017.7(6).127)

Foreign Literatures:

3. **Data-Driven Dynamic Pricing and Ordering with Perishable Inventory in a Changing Environment**

Description / Summary (Foreign Literature). Keskin, Li, and Song examined pricing and inventory decisions for perishable products when retailers do not have complete knowledge of demand, price relationships, and perishability. Their work also considers how the relationship between demand and price may change over time. This supports the proposed study's use of historical sales, pricing, inventory, and perishability information when evaluating pricing decisions.

Source: <https://doi.org/10.1287/mnsc.2021.4011>

4. **Dynamic Pricing for Perishable Goods: A Data-Driven Digital Transformation Approach**

Description / Summary (Foreign Literature). Syed et al. examined how supermarkets can implement data-driven dynamic pricing for products with short shelf lives. The study highlights the importance of reliable data collection, analytics, algorithms, and continuous adjustment of pricing strategies. It supports the proposed system's use of retail data and analytical models to provide pricing recommendations for perishable products.



Source: <https://doi.org/10.1016/j.ijpe.2024.109405>

5. **Deep Reinforcement Learning Algorithms for Dynamic Pricing and Inventory Management of Perishable Products**

Description / Summary (Foreign Literature). This study explains that perishable products create additional pricing and inventory challenges because they have limited shelf lives. It also shows that product shelf life increases the complexity of optimization because inventory conditions change as products age. This supports including remaining shelf life and inventory as important variables in the proposed pricing framework.

Source: <https://doi.org/10.1016/j.asoc.2024.111864>

6. **A New Dynamic Pricing Model for the Effective Sustainability of Perishable Product Life Cycle**

Description / Summary (Foreign Literature). Tekin and Erol (2017) argue that much of the perishable stock that supermarkets throw away is caused by pricing policies that ignore product age, not only by over-ordering. They develop a pricing policy for stock whose shelf life is about to expire, so that it still sells against fresher stock, with the aims of reducing waste cost and increasing supermarket profit.

Relevance to the proposed study. It directly supports using remaining shelf life as an input to pricing, which is the core of the proposed price optimization for milk products.

Source: <https://doi.org/10.3390/su9081330>

Review of Related Studies (RRS/s)

The RRS focuses on existing research, systems, applications, or projects that address a problem similar to the one the group wants to solve.

Local Studies:

No Philippine study on demand forecasting or dynamic pricing for milk retail was found in the group's search, which is part of the gap this study addresses.

Foreign Studies:

1. **Dynamic Pricing for Perishable Goods: A Data-Driven Digital Transformation Approach**

Description / Summary (Foreign Study). Syed et al. conducted a multi-case study on dynamic pricing in supermarkets. The researchers developed a three-phase framework covering data preparation, analytical implementation, and continued pricing adaptation. The study shows that dynamic pricing for perishable goods requires more than an algorithm because data quality, implementation, and changing customer behavior also affect the pricing process.

Relevance to the proposed study. It supports developing a data-driven pricing tool designed specifically for perishable retail products rather than a general retail forecasting system.

Source: <https://doi.org/10.1016/j.ijpe.2024.109405>

2. **Data-Driven Dynamic Pricing and Ordering with Perishable Inventory in a Changing Environment**

Description / Summary (Foreign Study). Keskin, Li, and Song used real-world data from a major supermarket chain to investigate joint pricing and inventory decisions. Their study considered uncertainty in the demand-price relationship, demand variation, inventory perishability, and changes in customer response to price over time. They also extended their approach to account for age-dependent perishability.

Relevance to the proposed study. It provides support for considering price, demand, inventory, and product age together instead of optimizing prices using sales history alone.

Source: <https://doi.org/10.1287/mnsc.2021.4011>



3. **Deep Reinforcement Learning Algorithms for Dynamic Pricing and Inventory Management of Perishable Products**

Description / Summary (Foreign Study). The researchers studied dynamic pricing and inventory management as a combined problem for products with limited shelf lives. They used deep reinforcement learning because conventional optimization becomes increasingly difficult as the number of product ages, inventory conditions, and pricing decisions increases.

Relevance to the proposed study. The research demonstrates that artificial intelligence can support pricing decisions for perishable inventory. However, the proposed study will examine more accessible supervised machine learning models such as Linear Regression, Random Forest, and XGBoost for demand forecasting before performing price optimization.

Source: <https://doi.org/10.1016/j.asoc.2024.111864>

4. **Augmented Data and XGBoost Improvement for Sales Forecasting in the Large-Scale Retail Sector**

Description / Summary (Foreign Study). Massaro, Panarese, Giannone, and Galiano (2021) applied XGBoost in an industrial project for a large-scale retailer to predict product sales, including under promotion conditions. Because the available data were not enough to reach the accuracy they needed, they trained and tested the model with an augmented data technique.

Relevance to the proposed study. It shows XGBoost, the primary model named in the title, working on retail sales forecasting, and it offers a way forward if the partner store's sales history turns out to be short.

Source: <https://doi.org/10.3390/app11177793>

5. **Analytics for an Online Retailer: Demand Forecasting and Price Optimization**

Description / Summary (Foreign Study). Ferreira, Lee, and Simchi-Levi (2016) worked with the online retailer Rue La La, which prices limited-time sales of products it has never sold before. They used machine learning to estimate lost sales and forecast demand, then turned those forecasts into daily pricing decisions through an optimization algorithm.

Relevance to the proposed study. It is a model for the study's two-stage design, in which a demand forecast feeds a price optimization step, applied here to milk products with a remaining shelf-life constraint.

Source: <https://doi.org/10.1287/msom.2015.0561>

Research and System Gap

The reviewed sources show that pricing perishable products is harder than general retail pricing, because inventory ages, the relationship between price and demand shifts over time, and unsold stock becomes waste. Existing approaches range from data-driven pricing frameworks to deep reinforcement learning.

The gap. The reviewed work comes from large foreign retailers and uses methods that are complex to deploy. It does not offer Philippine small and medium retailers an accessible tool that forecasts milk demand with supervised models and turns remaining shelf life into a pricing recommendation.

Refined Proposed Title and Solution

Demand Forecasting and Dynamic Price Optimization for Milk Retail Products Considering Remaining Shelf Life Using XGBoost (16 words). The title now names XGBoost, the primary model among the three compared, because it handles the nonlinear effects of price, inventory, and remaining shelf life and reports feature importance. "Machine Learning-Based" was removed, since naming the algorithm already says it, which keeps the title within 16 words. Linear Regression and Random Forest remain as the comparison models.

References for the RRL and RRS

Barrion, A. S. A., Calayag, J. A. S., Nguyen-Orca, M. F. R., & Melo-Rijik, M. N. (2023). Food loss and waste in the Philippines: A literature review. *Food Research*, 7(6), 278-289. [https://doi.org/10.26656/fr.2017.7\(6\).127](https://doi.org/10.26656/fr.2017.7(6).127)



- Ferreira, K. J., Lee, B. H. A., & Simchi-Levi, D. (2016). Analytics for an online retailer: Demand forecasting and price optimization. *Manufacturing & Service Operations Management*, 18(1), 69-88.
<https://doi.org/10.1287/msom.2015.0561>
- Inokuma, M., Ikeda, S., & Takano, Y. (2026). Estimating profitable price bounds for prescriptive price optimization. *Discover Analytics*, 4, Article 21. <https://doi.org/10.1007/s44257-026-00072-7>
- Keskin, N. B., Li, Y., & Song, J.-S. (2022). Data-driven dynamic pricing and ordering with perishable inventory in a changing environment. *Management Science*, 68(3), 1938-1958. <https://doi.org/10.1287/mnsc.2021.4011>
- Liu, C., & Sustik, M. A. (2021). Elasticity based demand forecasting and price optimization for online retail. *arXiv*.
<https://arxiv.org/abs/2106.08274>
- Massaro, A., Panarese, A., Giannone, D., & Galiano, A. (2021). Augmented data and XGBoost improvement for sales forecasting in the large-scale retail sector. *Applied Sciences*, 11(17), 7793.
<https://doi.org/10.3390/app11177793>
- Syed, T. A., Aslam, H., Bhatti, Z. A., Mehmood, F., & Pahuja, A. (2024). Dynamic pricing for perishable goods: A data-driven digital transformation approach. *International Journal of Production Economics*, 277, 109405.
<https://doi.org/10.1016/j.ijpe.2024.109405>
- Tekin, P., & Erol, R. (2017). A new dynamic pricing model for the effective sustainability of perishable product life cycle. *Sustainability*, 9(8), 1330. <https://doi.org/10.3390/su9081330>
- U.S. Department of Agriculture Foreign Agricultural Service. (2024). Philippines: Dairy and products annual (Report No. RP2024-0038).
https://apps.fas.usda.gov/newgainapi/api/Report/DownloadReportByFileName?fileName=Dairy+and+Products+Annual_Manila_Philippines_RP2024-0038.pdf
- XGBoost Developers. (n.d.). XGBoost parameters. XGBoost Documentation.
<https://xgboost.readthedocs.io/en/latest/parameter.html>
- Yavuz, T., & Kaya, O. (2024). Deep reinforcement learning algorithms for dynamic pricing and inventory management of perishable products. *Applied Soft Computing*, 163, 111864.
<https://doi.org/10.1016/j.asoc.2024.111864>



FINAL COMPARISON OF THE FIVE TITLES

Rated 1 to 5, where 5 is highly feasible and 1 is not currently feasible

Criteria	Title 1 BAPTC	Title 2 SM Foundation	Title 3 QC ITDD	Title 4 Immunization	Title 5 Milk Retail
Technical and technology feasibility	4	4	4	5	5
Dataset and data feasibility	3	3	3	3	3
Algorithm or model feasibility	4	5	4	4	4
Operational feasibility	4	4	4	3	3
Economic feasibility	4	5	5	5	5
Schedule feasibility	3	3	3	4	4
Legal and ethical feasibility	4	4	3	4	4
Risk manageability	4	4	4	4	3
Total score /40	30	32	30	32	31
Group's ranking	1	4	2	5	3

Reading the comparison. The five totals sit between 30 and 32, a narrow spread that shows none of the titles is unworkable. Titles 2 and 4 score highest, at 32 each. Titles 1 and 3 score lowest, at 30 each, and for the same reason: each waits on an institution to release records. In both cases the request has already reached the right office and has been acknowledged, so what is outstanding is a reply and the shape of the records, not a decision on whether to cooperate.

Why the ranking is not simply the score order. The scores measure how safely a title can be delivered, not what the study contributes or whom it serves. Title 1 is placed first because it remains the group's committed direction, because BAPTC and DA-CAR have both acknowledged the group's requests, and because it addresses a Philippine market that no publicly identified study has modelled for price and supply glut together. Title 3 is placed second and kept as the backup title, because it is the direction the professor responded to most strongly and because its method can be validated on public benchmark data while the city's reply is pending. Title 5 is placed third because the group is more certain that it can secure both a dataset and a partner retailer for it, and because it does not depend on an institution agreeing to release records. Title 2 is placed fourth. The group has a closer contact at SM Foundation and reaching the client is easier than for most titles, but the partnership is still uncertain. Title 4 is placed fifth, below Title 2, because child immunization records are health data, which in the Philippines are highly confidential and among the hardest records to obtain. The group is still looking for a partner health center, and reaching 10,000 records from a single center is itself difficult.

FINAL RECOMMENDATION

The group's ranking of the titles, justified from each title's RRL and RRS

How the ranking was decided. The group ranked the titles itself. The feasibility scores in the comparison table measure how safely each title can be delivered, but the ranking also weighs which titles the group is committed to, which partner the group can realistically secure, how sensitive and obtainable each dataset is, and how strongly each title's review of related literature and studies supports its problem and its method. Each title below is given with its role, the reason for its rank, what its RRL and RRS establish, the research gap it fills, and the condition under which the group would move to it. The group pursues the first title and moves down the list only when the title above it fails its data decision gate.



Rank 1: Primary Recommended Title	Highland Vegetable Price and Supply Glut Forecasting at the Benguet Agri-Pinoy Trading Center Using XGBoost
Partner and data status	BAPTC (Office of the General Manager) and DA-CAR AMAD. The formal request was sent on September 10, 2026 and acknowledged on September 14, 2026. BAPTC does not release raw or source files, so the request is for an aggregate extract of price and trading volume by commodity. DA-CAR AMAD replied that its concerned section will respond. DA Bantay Presyo bulletins and PAGASA-BSU weather records are public and already confirmed.
Why it ranks first	It is the group's committed direction, and both offices that hold the data have acknowledged the group's requests. Its research gap is the clearest of the five titles. The only BAPTC-specific price model found (Lubag et al., 2025) forecasts a single commodity's price and does not flag supply gluts, so a commodity-level early warning of price and glut risk does not yet exist for this market. The partner has already asked for this kind of tool. The Department of Agriculture (2021) frames farmer income at BAPTC as the problem to solve, and the Philippine Information Agency and OpenGov Asia both report DA-CAR moving BAPTC toward digital trading and artificial intelligence. The system fits a direction the agency has chosen. It serves an identifiable group of beneficiaries, the farmers of Benguet, for whom a price collapse after an overlapping harvest is a direct loss. Its main risk is bounded. Its feasibility total of 30 is the lowest, tied with Title 3, because the study waits on an institution releasing records. That risk is controlled by the Week 3 decision gate below: if volume data is withheld, the title narrows to price forecasting on public data instead of failing.
What the RRL establishes	Local literature. The Department of Agriculture (2021) describes BAPTC as the country's biggest vegetable trading facility. The Philippine Information Agency and OpenGov Asia confirm that digital market information at BAPTC is a live government priority. DA Bantay Presyo gives an independent public price series for cross-checking and for a price-only fallback. The approved FOI request for BAPTC programme documents shows what is publicly obtainable and supports framing the request around aggregates rather than raw records. Foreign literature. Hyndman and Athanasopoulos (2021) are the basis for rolling-origin validation and for requiring every model to beat naive and seasonal-naive forecasts. Chen and Guestrin (2016) show that XGBoost handles missing values and supports both regression and classification, which the price and glut tasks need. Qiao, Kang, and Ahn (2023) found that domestic production drove most vegetable price swings in South Korea, which is why BAPTC volume is essential for the glut claim. Mao et al. (2022) show that models trained on normal periods can fail during shocks, so error is reported by season. Haselbeck et al. (2022) found that machine learning outperformed classical forecasting on horticultural sales.
What the RRS establishes	Local studies. Lubag et al. (2025) forecast BAPTC lettuce prices with a bidirectional GRU (test R-squared of 0.95), which gives the study a benchmark but covers one commodity and no gluts. Quitaleg and Dumlao (2025) showed that Benguet production is forecastable and that supply-side data is the limiting factor. Hinlo and Lee (2026) found that demand explained 50 to 90 percent and supply shocks 9 to 50 percent of Philippine onion and garlic price variance, so price and supply have to be modelled together. Foreign studies. Li et al. (2024) reduced MAPE by 10 to 12 percent on 42 short-cycle vegetable products with a random forest and XGBoost model. Zeng, Gao, and Hu (2024) support a two-output design that forecasts price and volume separately. Nayak et al. (2024) found that weather inputs improved price forecasts, which is why PAGASA-BSU records are added. Paul et al. (2022) found that the best model differs by market, so XGBoost is benchmarked rather than assumed best. Bulut (2025) showed that trade volume makes forecasts more useful in practice.
Research gap it fills	Few studies forecast price and supply glut together, and none gives BAPTC or DA-CAR a commodity-level early warning of price movement and oversupply risk. The study delivers



	that warning as a staff-facing dashboard.
Technology and platform	Python with pandas, NumPy, statsmodels, scikit-learn, and XGBoost; PostgreSQL or SQLite; a Streamlit dashboard deployed locally or on a free tier. No external API is on the critical path.
Algorithm or model	XGBoost for price regression and glut-risk classification, compared against seasonal-naive, SARIMA, and random forest baselines under rolling-origin validation. Error is reported by commodity and season, with MAE, RMSE, and MAPE for price and precision, recall, F1, and PR-AUC for glut risk.
Dataset	BAPTC aggregate daily or weekly wholesale price and trading volume by commodity, still to be confirmed, joined with PAGASA-BSU daily rainfall and temperature. DA Bantay Presyo bulletins serve as the independent secondary series.
Cost and completion period	PHP 6,700 to 10,300 over an 18-week plan. Travel to La Trinidad is the dominant cost and can be reduced by using the remote request channel.
Major risks and mitigation	Unavailable or insufficient data, volume withheld while price is released, weak forecast accuracy, and schedule delay. Mitigated by the Week 3 gate, narrowing to price-only forecasting if volume is withheld, comparing every model against naive baselines, and building a minimum viable dashboard first.
Decision gate	Title 1 proceeds in full only if an aggregate of historical BAPTC price and trading-volume records, research-use permission, and adequate coverage are confirmed by Week 3. If price records arrive without volume, the title narrows to price forecasting and the supply-glut claim is withdrawn. If nothing is confirmed by Week 3, the group moves to Title 3.

Rank 2: Backup Title	A Data Governance System for Privacy Release Readiness of Quezon City Open Data Using Mondrian k-Anonymity
Partner and data status	Quezon City ITDD, which runs a documented request service, Control of Data Services in Conformity with the Data Privacy Act. The formal letter and email were sent on September 23, 2026, following an earlier request through an identified contact in the city. The reply and the shape of the record-level extract are pending.
Why it ranks second	It is the direction the professor responded to most strongly. It has the most robust fallback of the five titles. The method does not depend on one dataset, so it can be built and validated on public benchmark microdata while the city's reply is pending, and no development work waits on the city. It has the strongest legal grounding. Privacy is the subject of the study, not an obstacle to it, and the National Privacy Commission guidance that governs it has already been identified. It ranks below Title 1 because the group is committed to Title 1, and because this title's contribution still depends on a record-level extract whose shape is unconfirmed. As the backup, it is the title the group can switch to at the Week 3 gate without losing time.
What the RRL establishes	Local literature. Republic Act No. 10173 makes any record-level city dataset personal information until shown otherwise. Executive Order No. 2, s. 2016 requires government to disclose while protecting privacy, which is the tension the system resolves. NPC Advisory No. 2025-02 supports building privacy checks into the release process itself. The QC Open Data Dashboard confirms the partner, the candidate datasets, and that more releases are planned. The national portal, data.gov.ph, listed 175 datasets from 17 agencies and no local government publishers when the group checked it on September 19 and September 24, 2026. Foreign literature. Sweeney (2002) showed that records without names can still be re-identified, and defined k-anonymity. LeFevre, DeWitt, and Ramakrishnan (2006) proposed Mondrian, which keeps more detail than single-dimension methods at the same k. Machanavajhala et al. (2007) define the known weakness of k-anonymity, which the study reports as a limitation. Wang and Strong (1996) and Janssen, Charalabidis, and Zuidervijk (2012) show that data quality is multi-dimensional, and that privacy and quality concerns are documented reasons governments hold back data.



What the RRS establishes	Local studies. Perez (2025) analysed over 235,000 eFOI requests and found that about 44 percent contained personal information, even with privacy advisories in place. Perez (2020) found that agencies process requests inconsistently. Alampay (2021) and Pacis (2017) document the state of Philippine open data and the capacity barriers among government staff, and Pitogo (2019) found a national agency only partially compliant with the Data Privacy Act. Foreign studies. Rocher, Hendrickx, and de Montjoye (2019) estimated that 99.98 percent of Americans could be re-identified from 15 demographic attributes, and de Montjoye et al. (2013) identified 95 percent of 1.5 million people from four location and time points. El Emam et al. (2009) showed that the choice of k-anonymity algorithm changes how much detail survives. Vetrò et al. (2016) found that municipal releases without quality controls were measurably weaker. Prasser et al. (2020) describe ARX, which the group uses to cross-check its Mondrian output.
Research gap it fills	No published work applies these risk and quality measurements to a Philippine local government open-data release, and no Philippine study compares a smarter anonymisation method against the blanket suppression data owners actually use. The study delivers an auditable, reusable release-readiness procedure.
Method, cost, and period	Mondrian k-anonymity against fixed small-cell suppression as the baseline, with k reported across several values and ARX as a cross-check. An open-source Python stack. PHP 3,900 to 4,500 over 18 weeks, with travel as the only material cost.
Condition for moving to it	The group moves to Title 3 if Title 1's data conditions are not met by Week 3.

Rank 3: Second Alternative	Demand Forecasting and Dynamic Price Optimization for Milk Retail Products Considering Remaining Shelf Life Using XGBoost
Partner and data status	A partner retailer is still to be confirmed. The group is more certain of securing a retailer and its sales data for this title than for the partner-dependent titles below it, and public retail datasets are a fallback.
Why it ranks third	The group is more certain that it can secure both a dataset and a client for this title. It does not depend on a government office or foundation agreeing to release records, which is the risk that weighs on Titles 1, 2, and 3. It is the least expensive title, at PHP 1,000 to 4,000, and it involves no personal customer data. It ranks below Titles 1 and 3 because the group is committed to Title 1 and the professor favoured Title 3, and because its partner store is not yet confirmed and its price-optimization results depend on the store's prices having varied enough in the past.
What the RRL establishes	Local literature. The U.S. Department of Agriculture Foreign Agricultural Service (2024) reports that the Philippines imports about 99 percent of its dairy requirement, that local production supplies only about 20 percent of ready-to-drink milk, and that demand for it is forecast to grow. Milk is therefore a steady retail category made up largely of imported stock with a limited shelf life. Barrion et al. (2023) confirm that food waste is a recognised Philippine problem and that managing stock by age is an accepted way to reduce it. Foreign literature. Keskin, Li, and Song (2022) show that retailers do not fully know how demand responds to price or how fast stock perishes, and that this relationship changes over time. Syed et al. (2024) stress reliable data and continuous adjustment in perishable pricing. Yavuz and Kaya (2024) show that shelf life makes pricing and inventory decisions harder. Tekin and Erol (2017) argue that much perishable waste comes from pricing that ignores product age, which directly supports pricing by remaining shelf life.
What the RRS establishes	Foreign studies. Syed et al. (2024) developed a three-phase framework for dynamic pricing of perishables in supermarkets. Keskin, Li, and Song (2022) used a major supermarket chain's data and accounted for age-dependent perishability. Yavuz and Kaya (2024) used deep reinforcement learning, which the study replaces with more accessible supervised models. Massaro et al. (2021) applied XGBoost to retail sales forecasting and handled short data with augmentation. Ferreira, Lee, and Simchi-Levi (2016) turned demand forecasts into



	daily pricing decisions, which is the two-stage design this study uses. Local studies. No Philippine study on demand forecasting or dynamic pricing for milk retail was found, which is part of the gap.
Research gap it fills	The reviewed work comes from large foreign retailers and uses methods that are complex to deploy. It does not give Philippine small and medium retailers an accessible tool that forecasts milk demand with supervised models and turns remaining shelf life into a pricing recommendation.
Method, cost, and period	Linear Regression as the baseline against Random Forest and XGBoost, evaluated with MAE, RMSE, and MAPE. The selected model feeds a price optimization step that weighs predicted demand, cost, inventory, and remaining shelf life. PHP 1,000 to 4,000 over a 16-week plan.
Condition for moving to it	The group moves to Title 5 if neither Title 1 nor Title 3 can proceed.

Rank 4: Third Alternative	A Data Governance System for Verifiable CSR Impact Reporting Using Fellegi-Sunter Record Linkage
Partner and data status	SM Foundation. Dawn raised the partnership in person on September 21, 2026, the Foundation invited the proposal, and the letter was sent on September 23, 2026. Whether the Kabalikat sa Kabuhayan records are digital, whether names can be hashed on the Foundation's side, and whether a reviewed sample can be built are all unconfirmed.
Why it ranks fourth	The group has a closer contact at SM Foundation than at any other partner, so reaching the client is easier than for most titles. The partnership is still uncertain, however, and the record format and ground truth that decide the title are unconfirmed. It ranks above Title 4 because its data is less sensitive than child health records, the Foundation is both the data owner and the user, and an Exempt ethics route is achievable because the group never holds identifiable records.
What the RRL establishes	Local literature. Republic Act No. 11055 shows that the lack of a common identifier across records is a recognised Philippine problem, and explains why the study builds links from fields the Foundation already keeps rather than requesting PhilSys numbers. Foreign literature. Fellegi and Sunter (1969) give record linkage its decision-theoretic basis, which lets the system attach chosen error rates to the beneficiary count. Christen and Goiser (2007) supply the evaluation measures. Binette and Steorts (2022) frame the pipeline of blocking, comparison, and classification. Harron et al. (2017) argue that linkage quality must be reported alongside the result.
What the RRS establishes	Local studies. Du (2026), writing from the Philippine Statistics Authority, identifies duplicate records as a persistent national data quality problem. Zuniga et al. (2026) describe a Philippine health information exchange that needed a single identity record to deduplicate patients across facilities. Foreign studies. Enamorado, Fifield, and Imai (2019) show Fellegi-Sunter working on millions of records through fastLink. Li, Xu, and Grannis (2022) give a tested way to handle missing fields. Hand and Christen (2018) are the reason precision and recall are reported separately.
Research gap it fills	No published study applies probabilistic linkage to a Philippine corporate or NGO livelihood programme where no shared identifier exists, or measures linkage performance on Philippine name conventions. The Foundation's own published totals disagree, at 28,550, over 30,000, and over 32,000 graduates, which shows the problem is real.
Method, cost, and period	Fellegi-Sunter as the primary model, with exact matching and a Jaro-Winkler threshold as baselines. PHP 2,000 to 4,000 over 18 weeks, since the Foundation is in Metro Manila.
Condition for moving to it	The group moves to Title 2 if Titles 1, 3, and 5 cannot proceed and the Foundation confirms digital records and a reviewed sample.



Rank 5: Fourth Alternative	Development of a Fuzzy Matching and Data Quality Scoring Model for Detecting Potential Duplicate and Incomplete Child Immunization Records at a Barangay Health Center in Mandaluyong City
Partner and data status	A barangay health center in Mandaluyong City. The specific health center is still being identified, so no request has been sent yet.
Why it ranks fifth	Child immunization records are health data, which in the Philippines are highly confidential and among the hardest records to obtain. The title's own RRL confirms this: Republic Act No. 10173 and its Implementing Rules and Regulations require restricted, lawful handling of these records and organisational, physical, and technical safeguards. The group is still looking for a partner health center, and reaching 10,000 records from a single center is itself difficult. Whether simulated records may count toward that floor is still to be confirmed with the professor. It ranks below Title 2 because its data is far more sensitive and its client has not yet been found, while SM Foundation has already invited a proposal.
What the RRL establishes	Local literature. Republic Act No. 10173 and its Implementing Rules and Regulations set the safeguards for child health data. The Universal Health Care Act and the Philippine eHealth Strategic Framework establish why reliable immunization information matters for service continuity and local reporting. Foreign literature. The WHO Immunization Agenda 2030 stresses using good information to find missed children. The WHO Data Quality Review Toolkit, Weiskopf and Weng (2013), and Kahn et al. (2016) define the completeness, consistency, and plausibility checks that become the tool's validation rules.
What the RRS establishes	Local studies. Acacio Claro et al. (2024) found that installing a system does not ensure consistently entered records. Conjares et al. (2025) document fragmented data management in Philippine health information systems. Elepano et al. (2025) found that privacy design and workflow fit decide whether primary care staff accept electronic records. Foreign studies. Sweeney et al. (2013) support fuzzy matching to flag possible duplicates rather than merge them automatically. Lewis et al. (2023) found no single standard method for assessing health record quality, and Jin et al. (2023) found missing, inconsistent, and duplicated data to be recurring problems.
Research gap it fills	The reviewed frameworks explain good data quality and immunization monitoring, but do not give a small Philippine barangay health center an explainable tool for flagging incomplete and possibly duplicate child immunization records.
Method, cost, and period	Rule-based validation, fuzzy duplicate matching, and a weighted Data Quality Score on de-identified or simulated records. PHP 1,500 to 5,500 over a 12-week plan.
Condition for moving to it	The group moves to Title 4 only if the titles above it cannot proceed and a health center gives written approval, or the professor accepts simulated records toward the 10,000-record floor.

Library Activity Documentation

Students should also document their research/library activities by inserting photographs in the three designated spaces provided in the form.

Picture during Library Activities:



Picture 1: Insert snapshot here



Picture 2: Insert snapshot here



Picture 3: Insert snapshot here





1. AI Interaction Log (if applicable)

If your group used any Generative AI tool while completing the assessment, document how the tool was used. The log should demonstrate that AI supported the group's work rather than replacing the group's research, analysis, and decision-making.

Group Member	AI Tool Used	Prompt / Purpose	AI Output Used For	How the Output Was Verified or Modified
Dawn Andrei Pamesa	Claude	Helped draft, organize, and format the revised feasibility studies and the RRL/RRS	Drafting assistance and document formatting	Every source was checked against its original publication or official page, and the content was reviewed and revised by the group
Rogelio Porras	ChatGPT	Generated search keywords for related literature and studies on vegetable price forecasting	Research keyword generation and research assistance	Sources were independently searched, read, and verified by the group
John Ashley Alba	ChatGPT	Generated search keywords for related literature and studies on health record data quality	Research keyword generation and research assistance	Sources were independently searched, read, and verified by the group
Ralph Blanco	ChatGPT	Generated search keywords for related literature and studies on perishable retail pricing	Research keyword generation and research assistance	Sources were independently searched, read, and verified by the group

2. AI Usage Declaration

We declare that any use of Generative AI in completing this Technical Assessment has been properly disclosed in the **AI Interaction Log**. We understand that AI is used only as an assistive tool and that our group remains responsible for verifying the accuracy, relevance, credibility, and originality of the submitted work.

We confirm that the final RRL/RRS, analysis, proposed title, and research outputs reflect our group's understanding and decisions. All literature, studies, and other external sources used have been properly cited and verified.

Group Name: Satisfied

Group Member	Signature
Dawn Andrei Pamesa	Dawn Andrei Pamesa
Rogelio Porras	Rogelio Porras
John Ashley Alba	John Ashley Alba
Ralph Blanco	Ralph Blanco

Date: September 24, 2026



3. Team Contribution Matrix

Complete the Team Contribution Matrix to document each member's participation in the assessment. Contributions should accurately reflect the work performed by each member.

Group Member	Assigned Task/Responsibility	Specific Contribution/Output	Contribution (%)
Dawn Andrei Pamesa	Group leader; Titles 2 and 3; partner outreach	Revised feasibility studies and RRL/RRS for Titles 2 and 3; partner letters; compiled and formatted the full paper	40%
Rogelio Porras	Title 1	RRL/RRS for the BAPTC title	20%
John Ashley Alba	Title 4	Revised feasibility study and RRL/RRS for the child immunization title	20%
Ralph Blanco	Title 5	Revised feasibility study and RRL/RRS for the milk retail title	20%
		Total	100%

Each member is expected to participate meaningfully in the completion of the assessment. The **total contribution percentage must equal 100%**.